

Graph Machine Learning

Alae Taoudi, Ouissal Boutouatou, Mohammed Sbahi

Paper Co-citation Prediction

March 2026

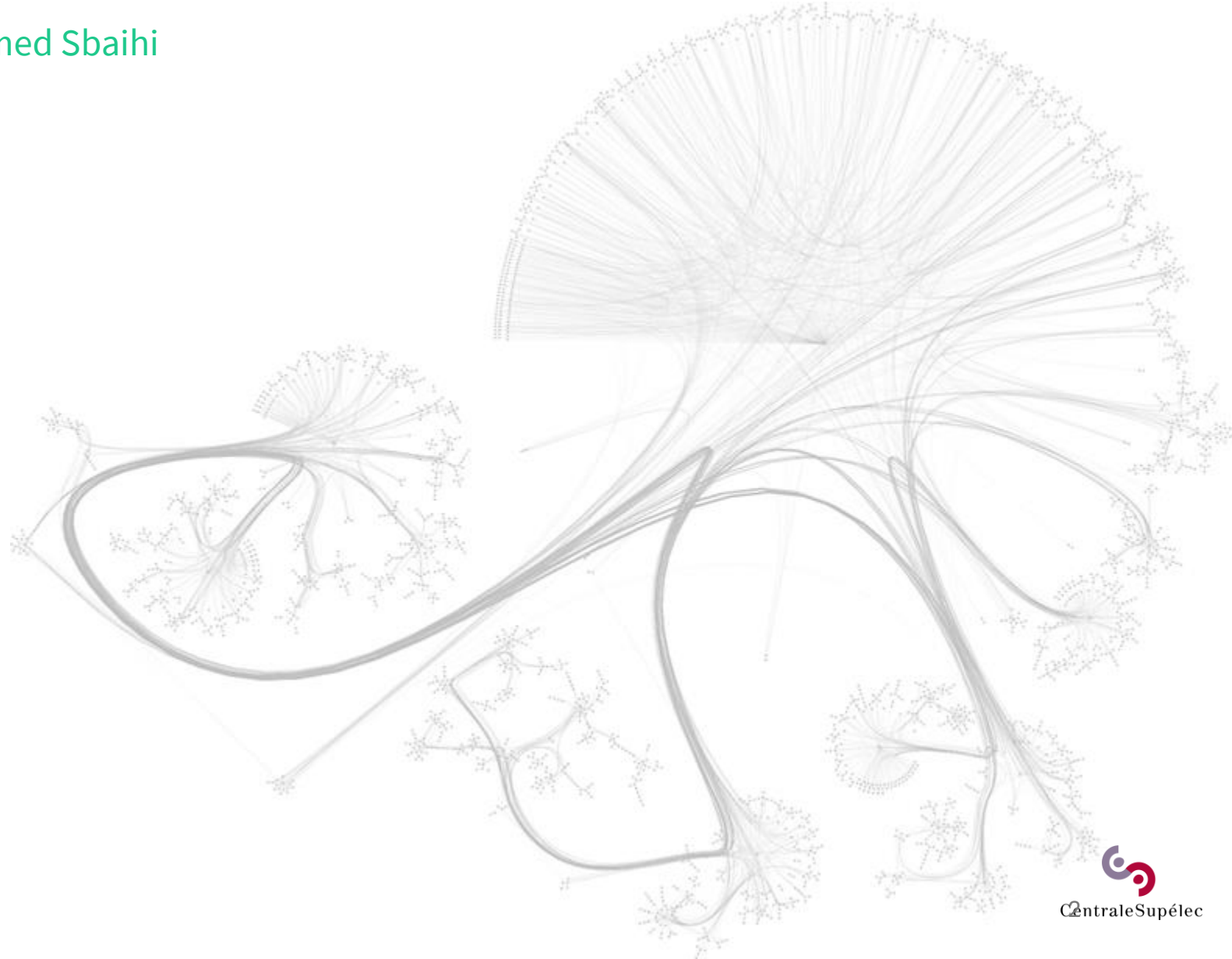


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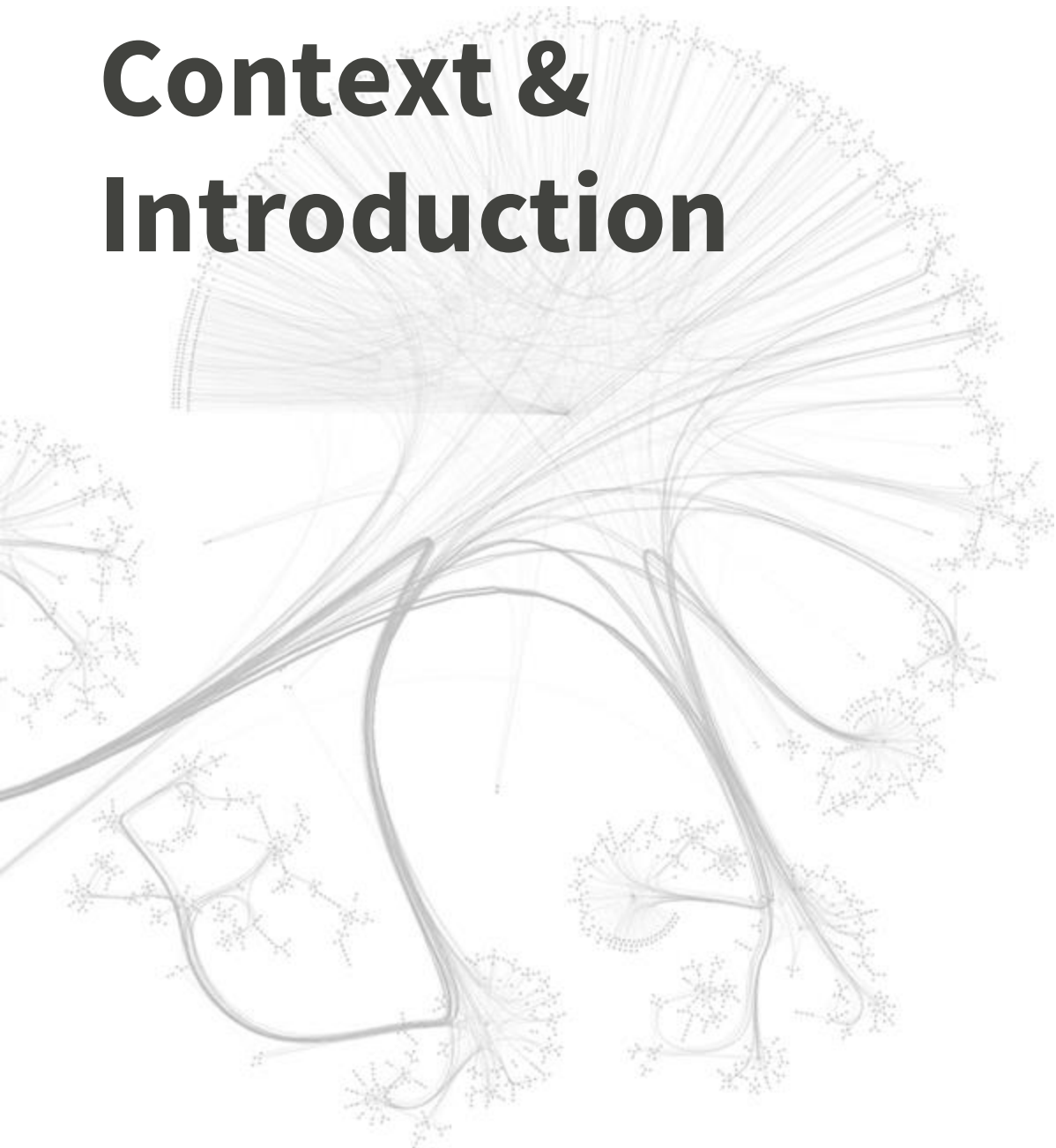
Presentation Plan

- Introduction & Context
- Graph Exploration
- Text Embeddings
- Node Representation
- Link Prediction
- Evaluation



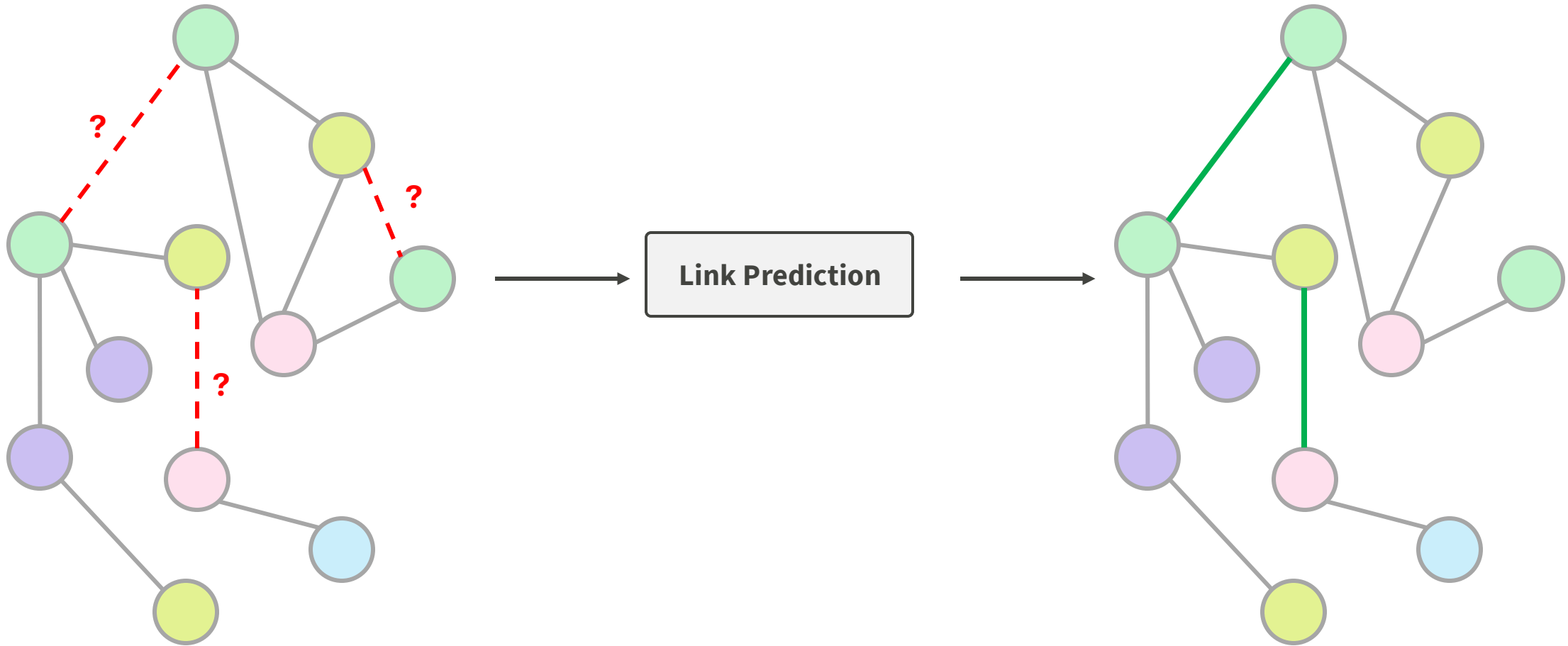
March 2026

Context & Introduction



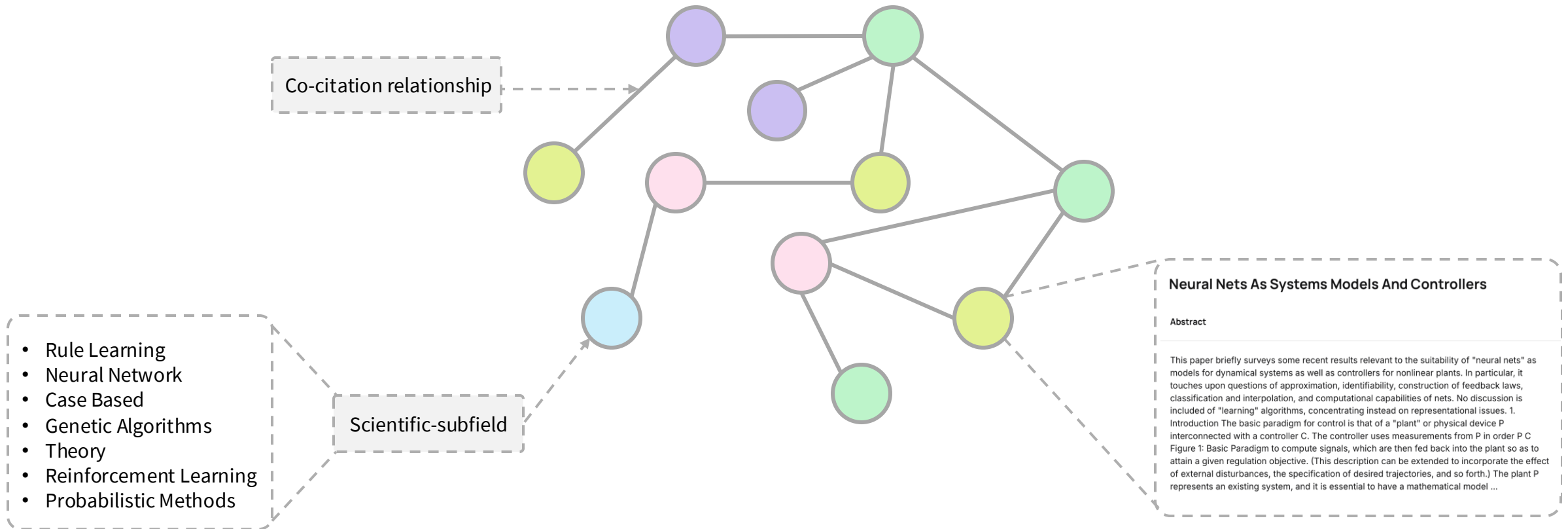
Task & **Graph**

Task – Link Prediction

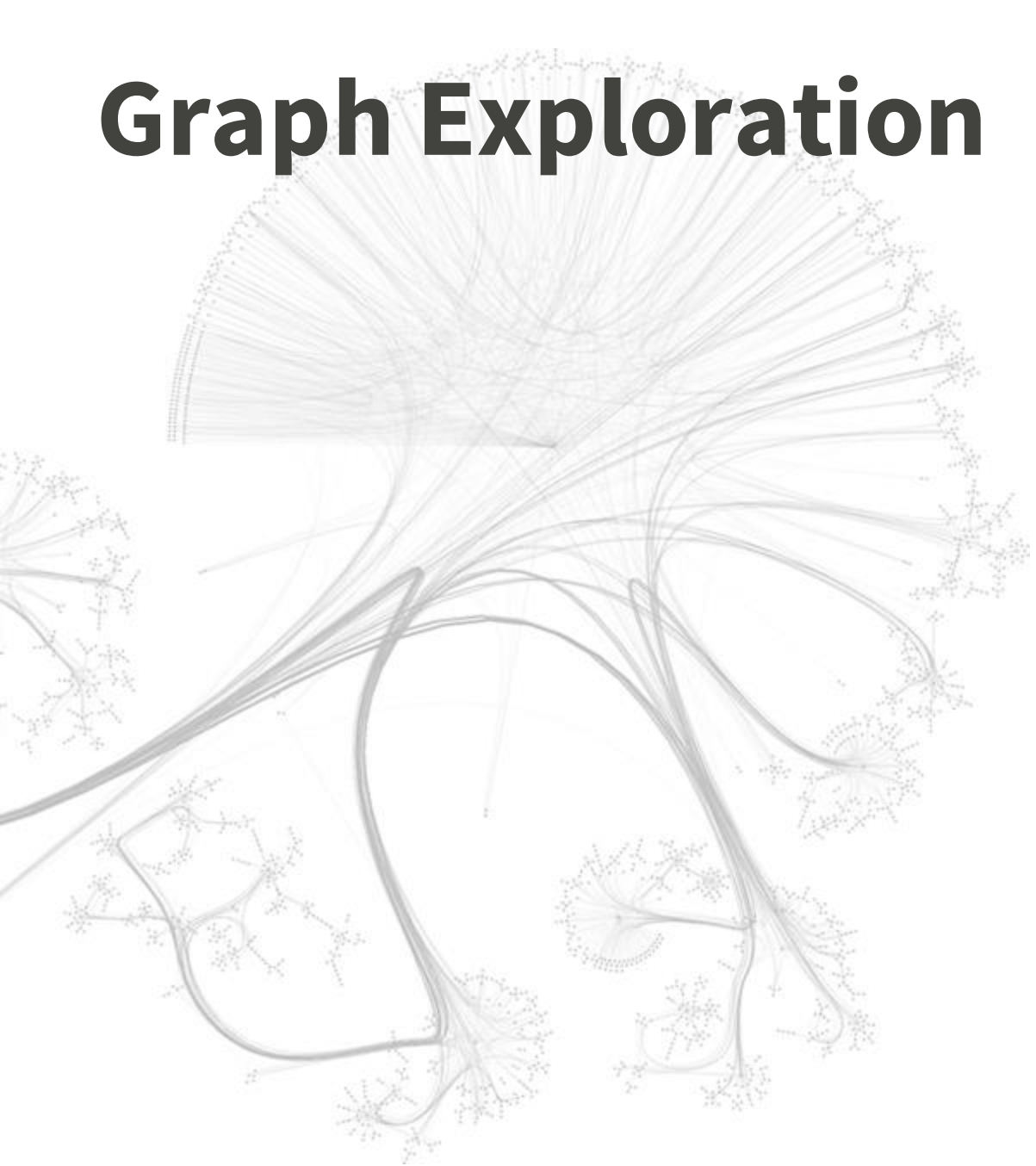


Subject Graph

CORA – co-citation graph



Graph Exploration

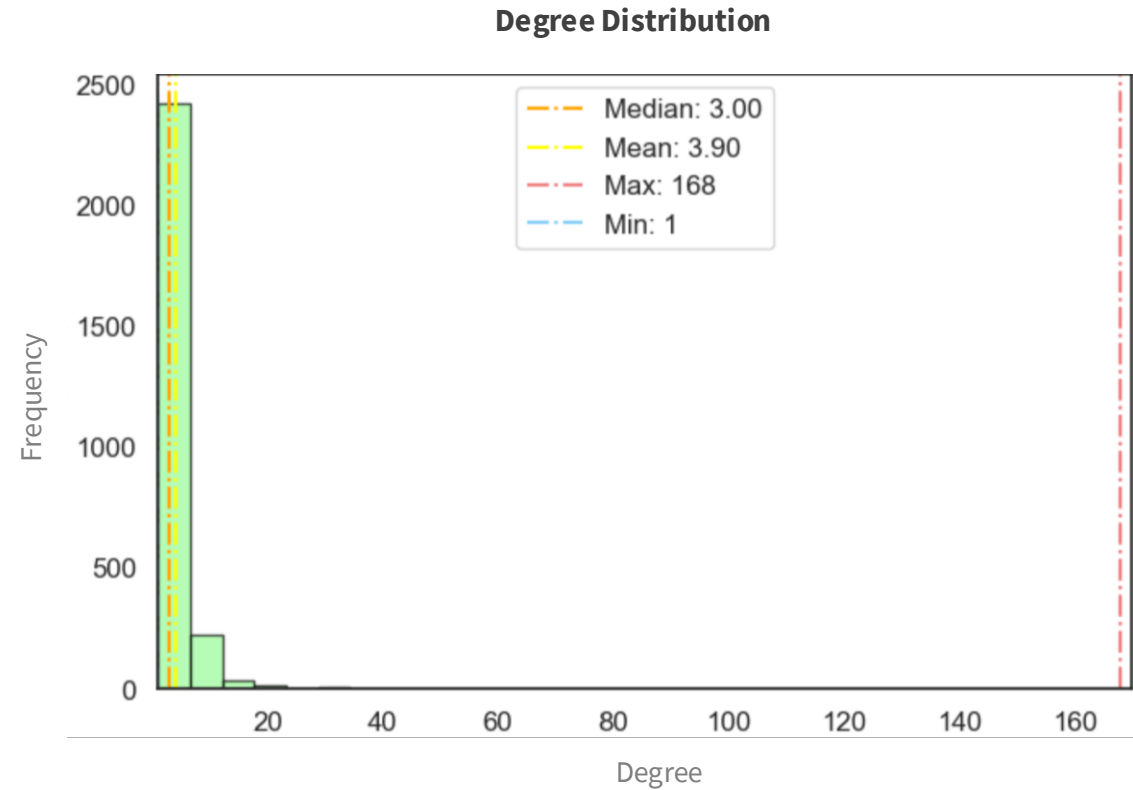


Descriptive & Homophily
Statistics

Graph Statistics

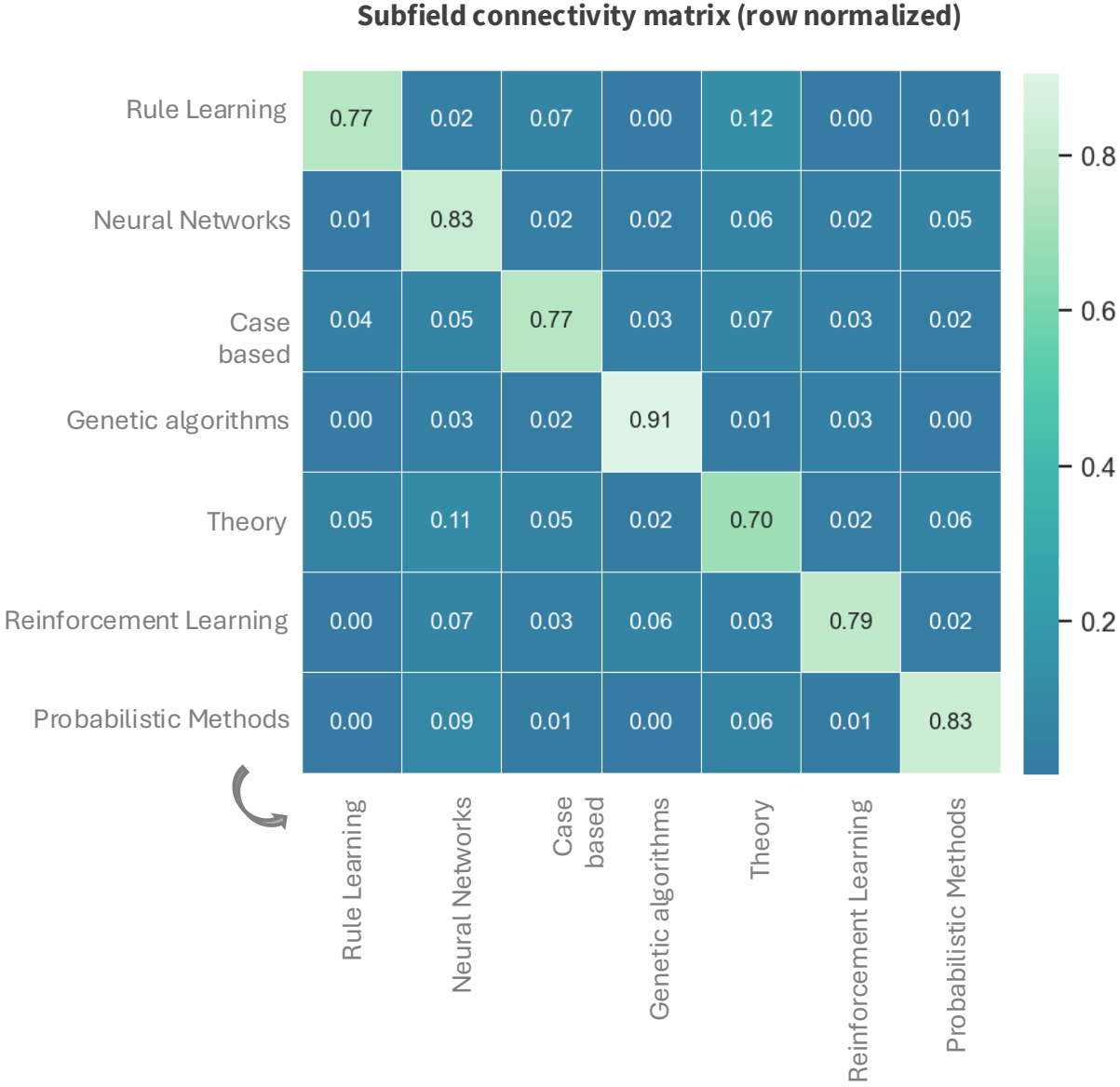
Descriptive Statistics

Nodes count	2708
Edges count	5278
Directed	False
Median degree	3

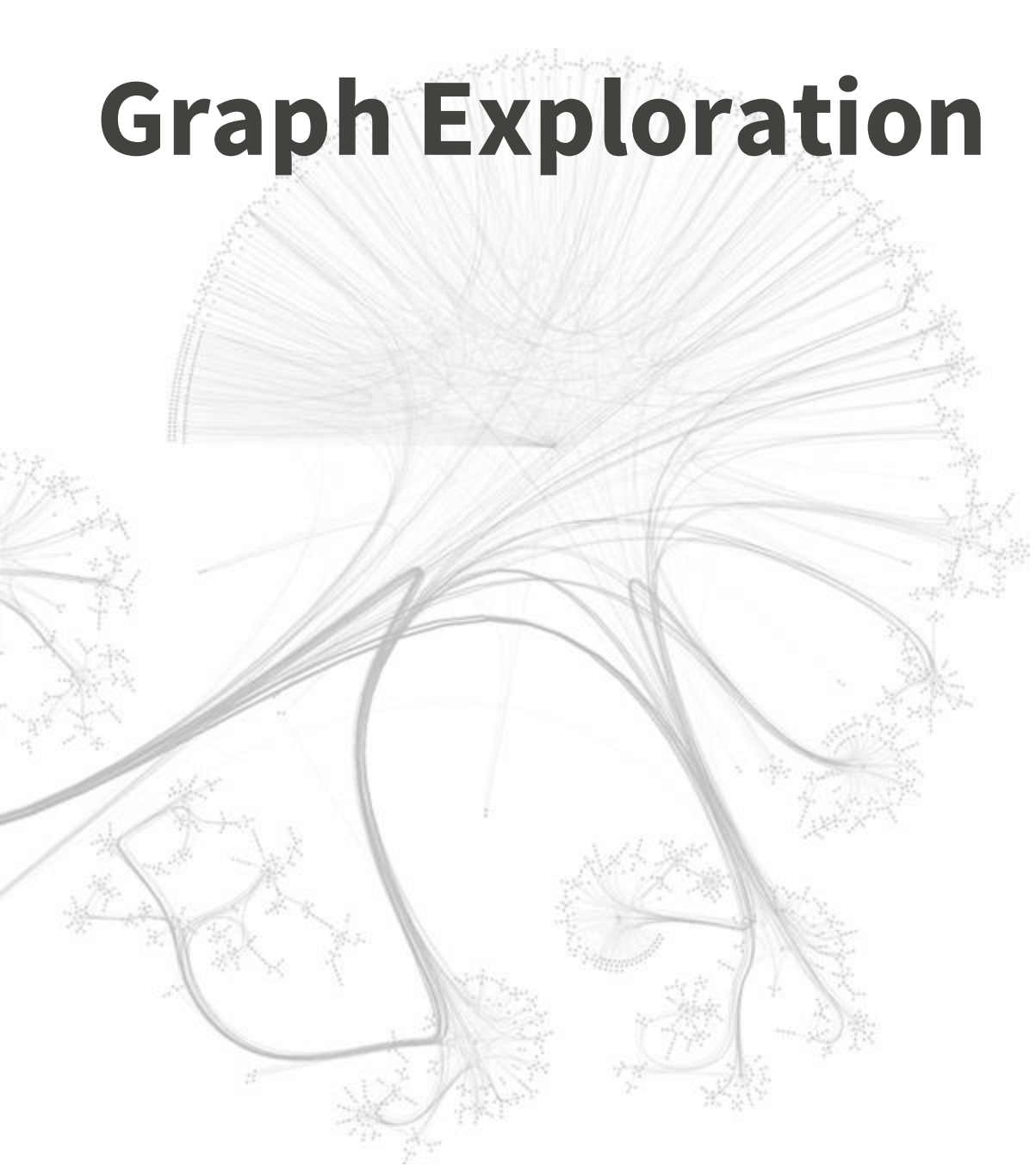


Homophily statistics

81% of all co-citation edges are within the same subfield

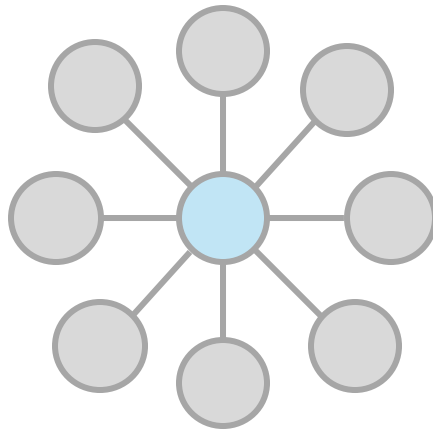


Graph Exploration



Centrality Measures

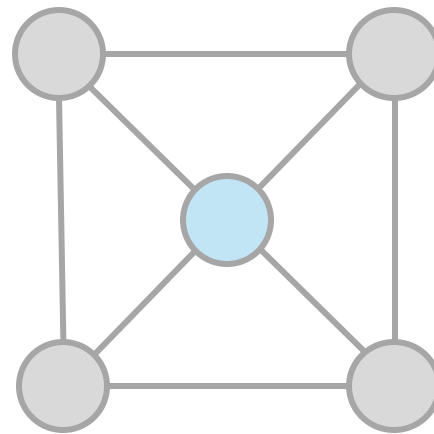
Influential nodes



Local popularity



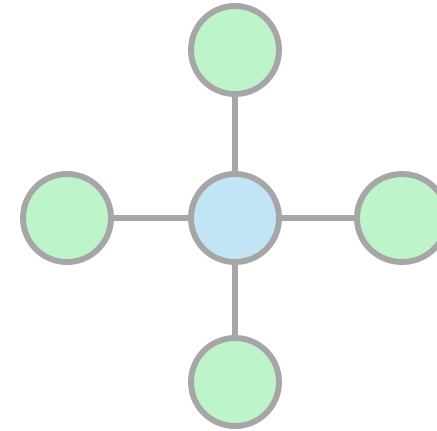
**Degree
centrality**



Community core



**Clustering
coefficient**



Global prestige



**Eigenvector
centrality**



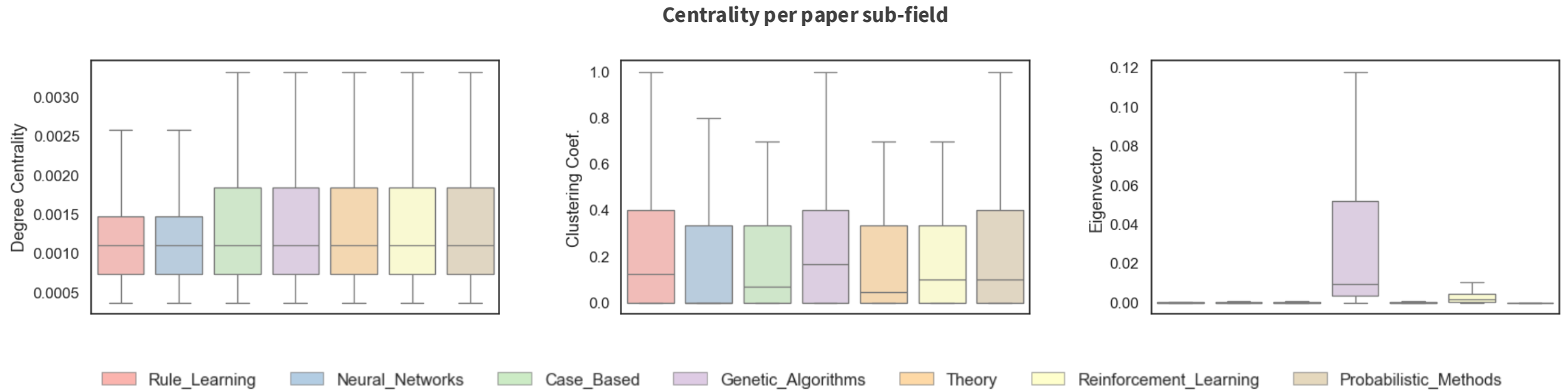
Centrality measures distribution



Scale-free network : few hubs dominate the co-citation landscape

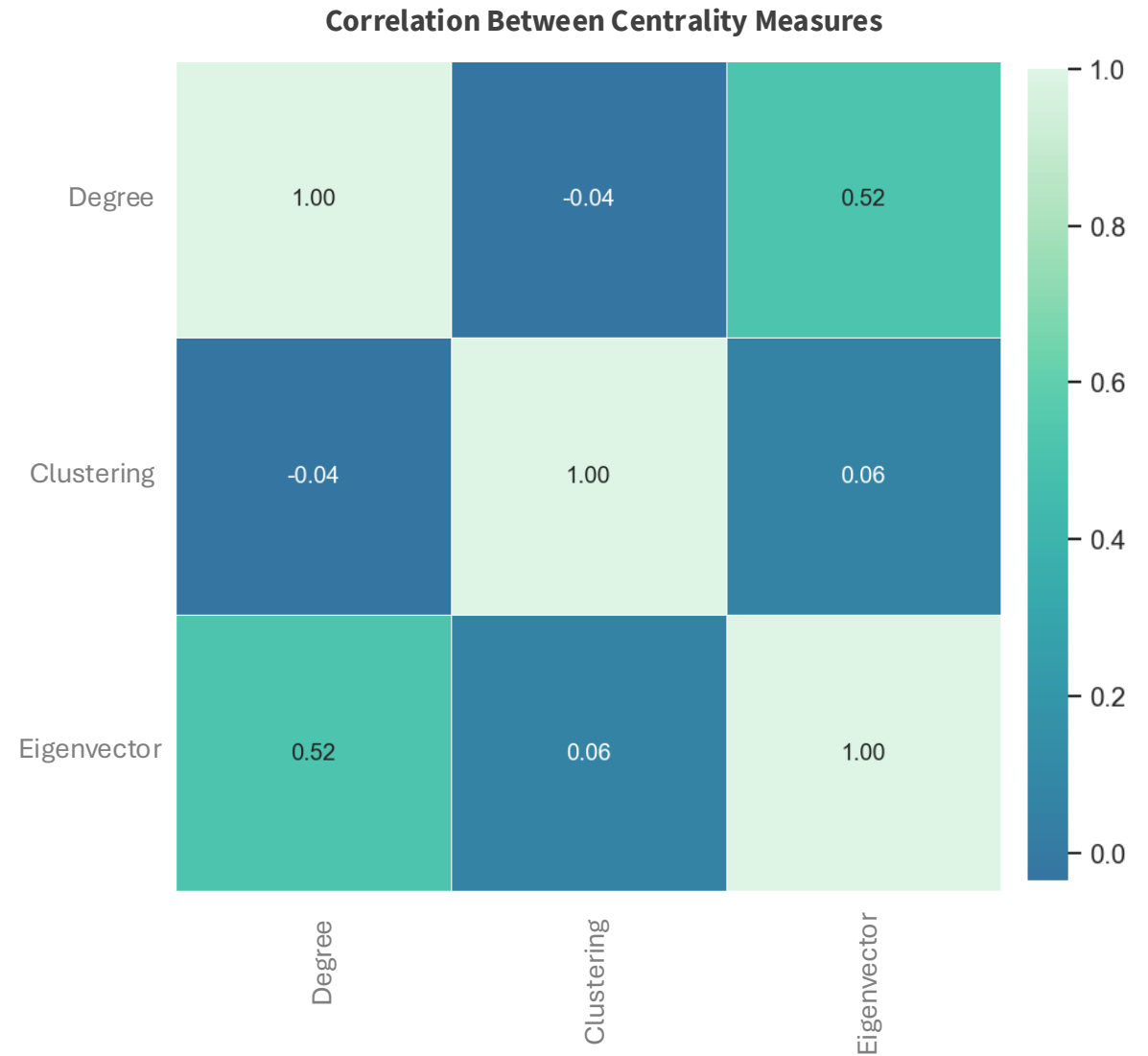


Centrality per sub-field

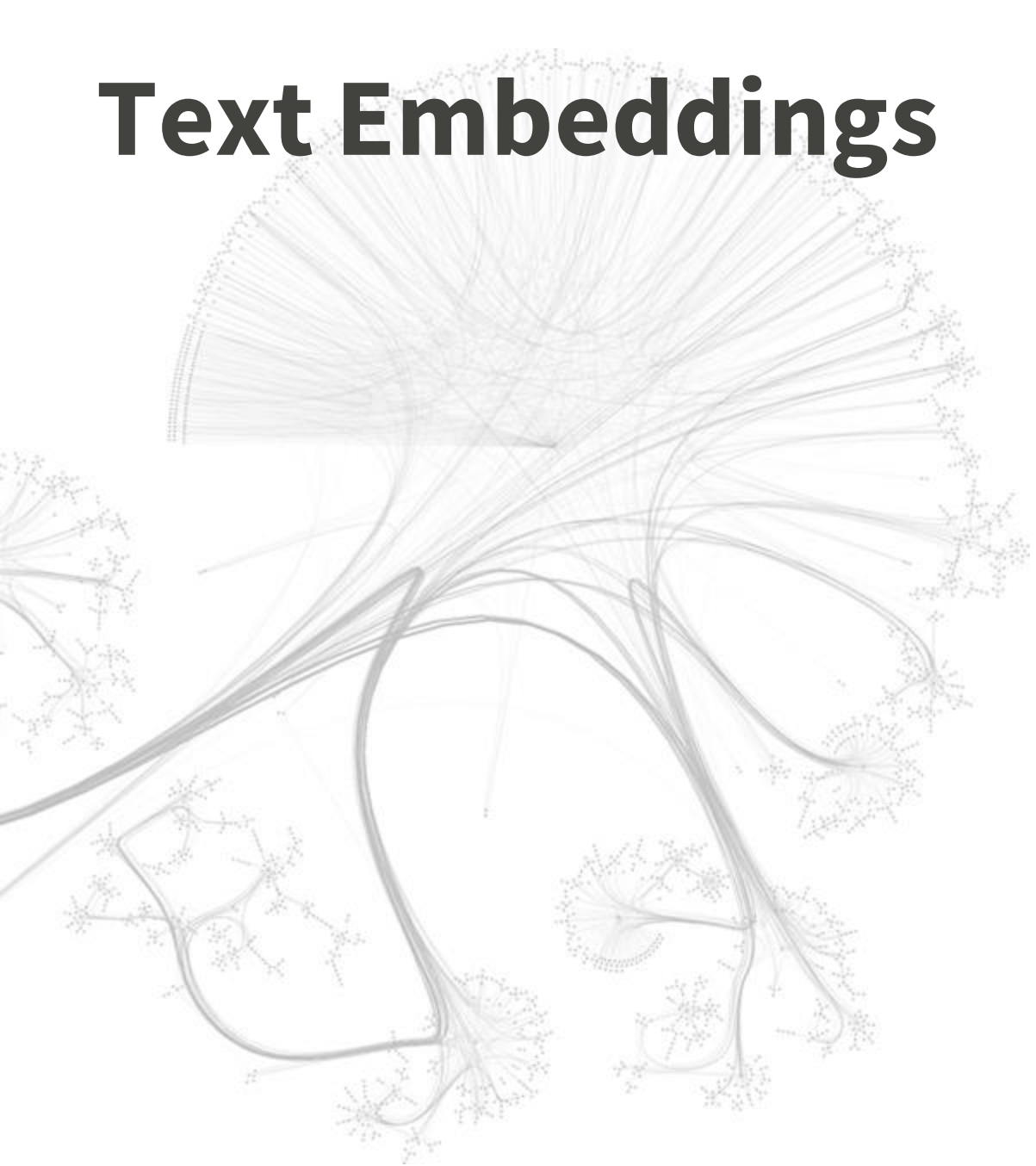


Centrality measures correlation

Centrality measures can make good candidates as node features for link prediction



Text Embeddings

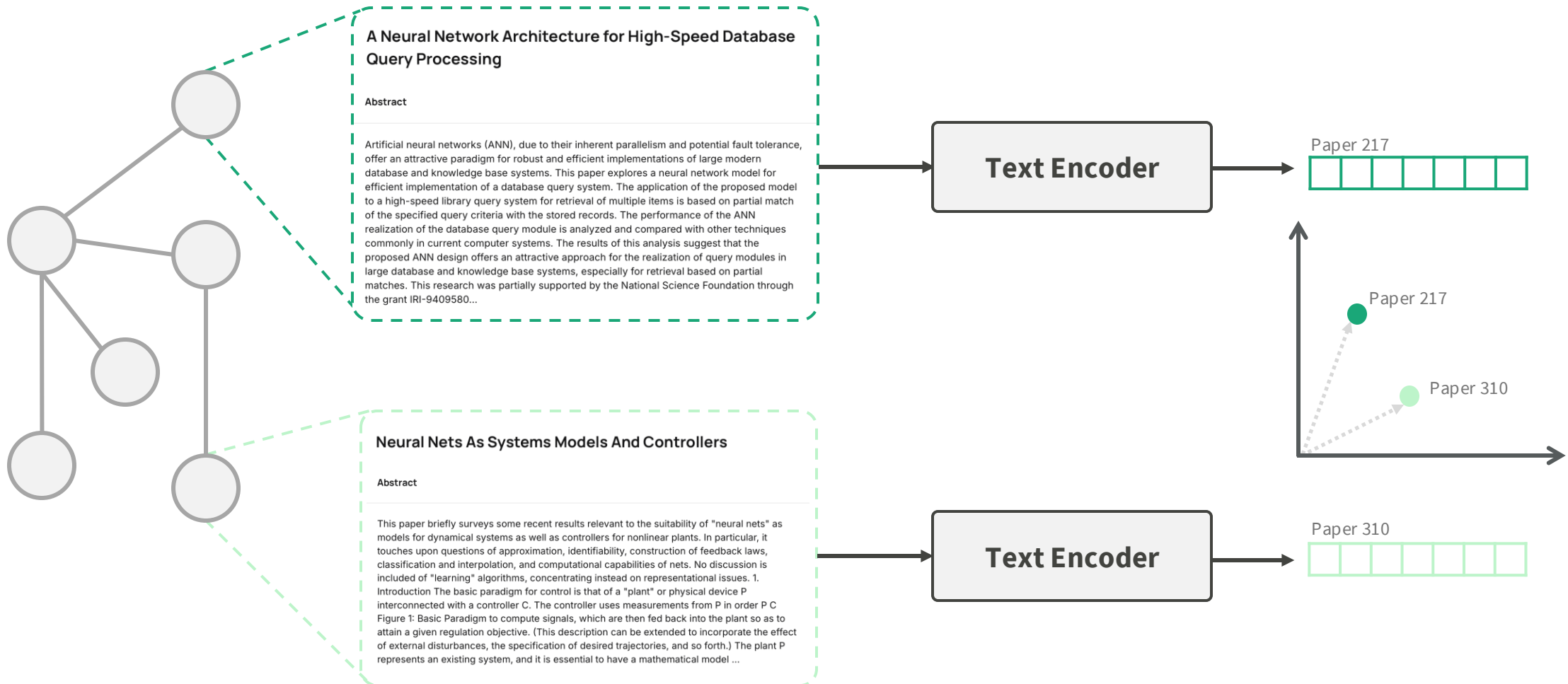


Attributes **Generation** from **Node**
Text Content



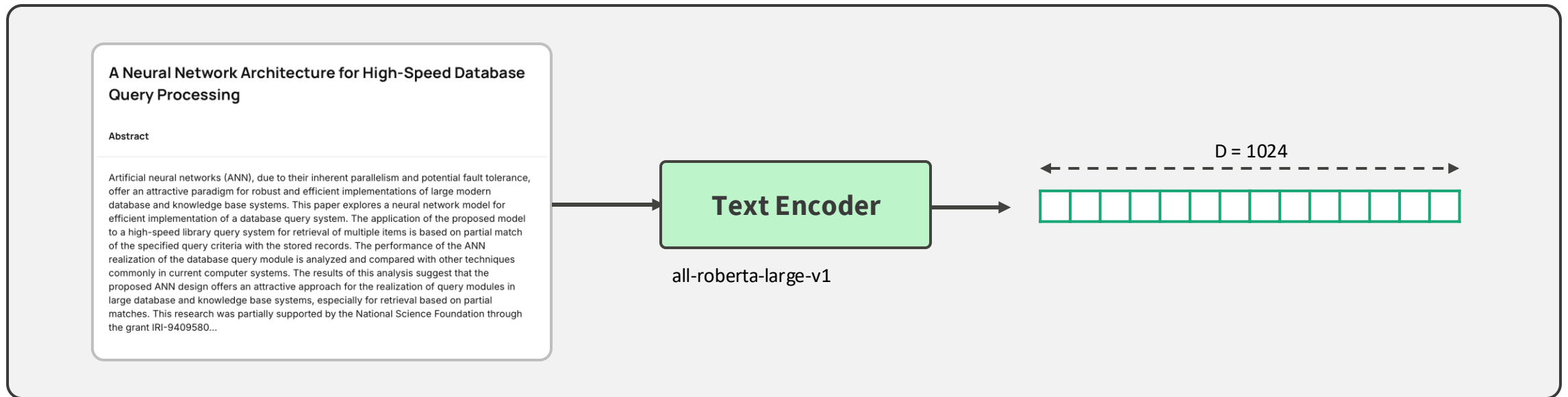
Text Embeddings

Attributes Generation from Node Text Content



Text Embeddings

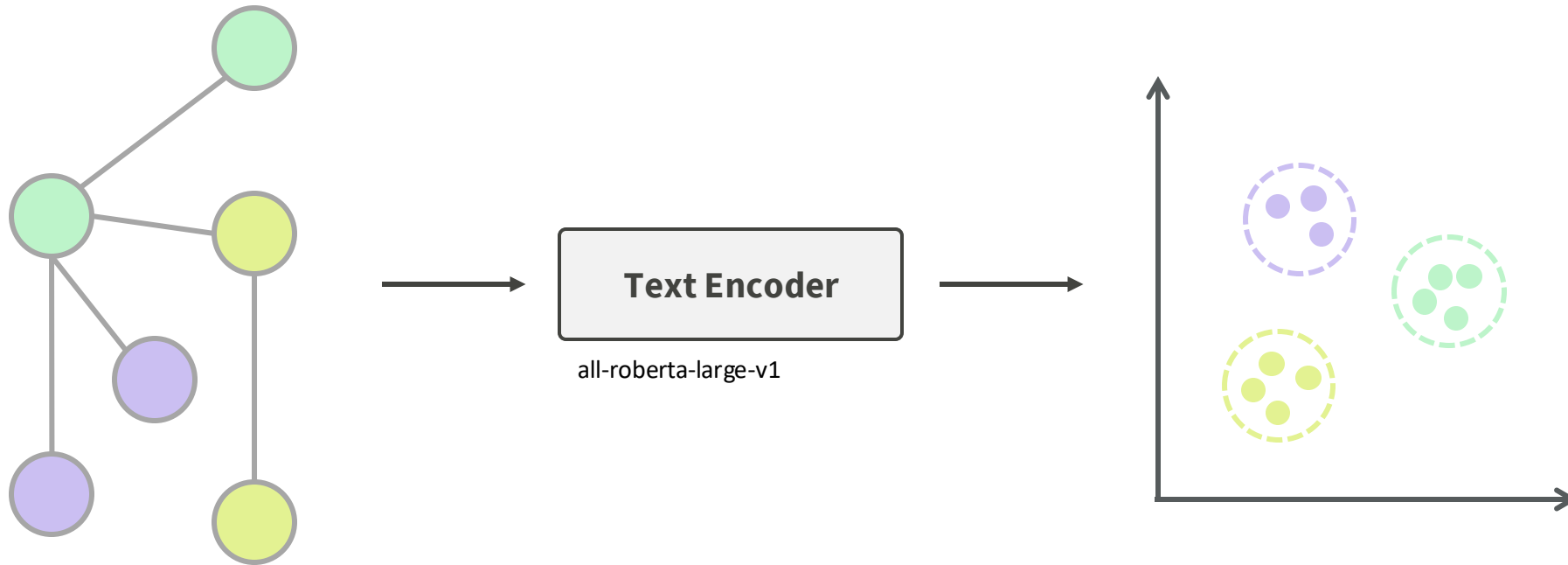
Methodology – Using a Pre-trained Language Representation Model



- How to assess the **quality** of the generated text embeddings?
- How well are text similarities and actual connections **aligned**?

Text Embeddings

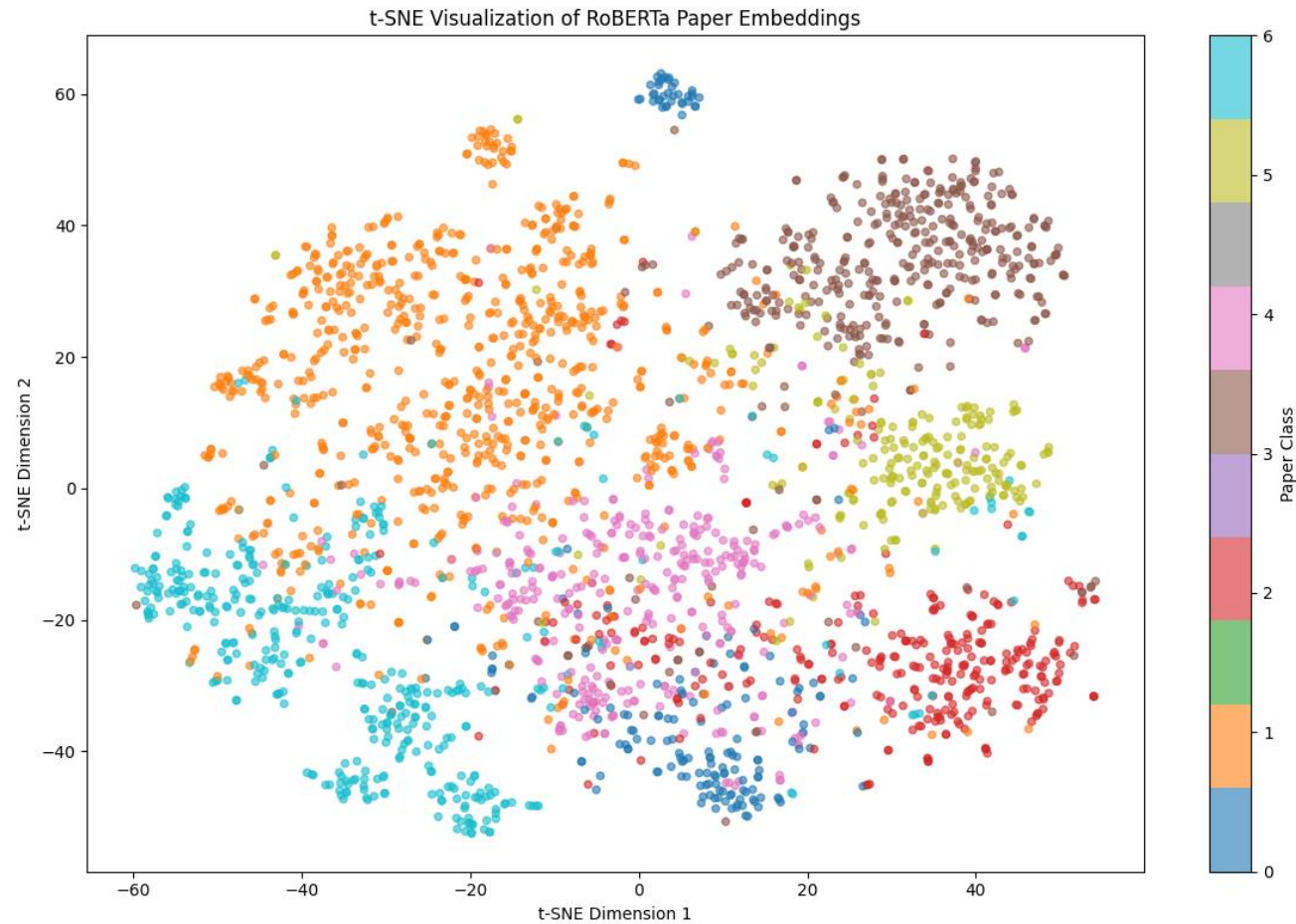
Text Embedding Quality Assessment



Are labels preserved in the latent space?

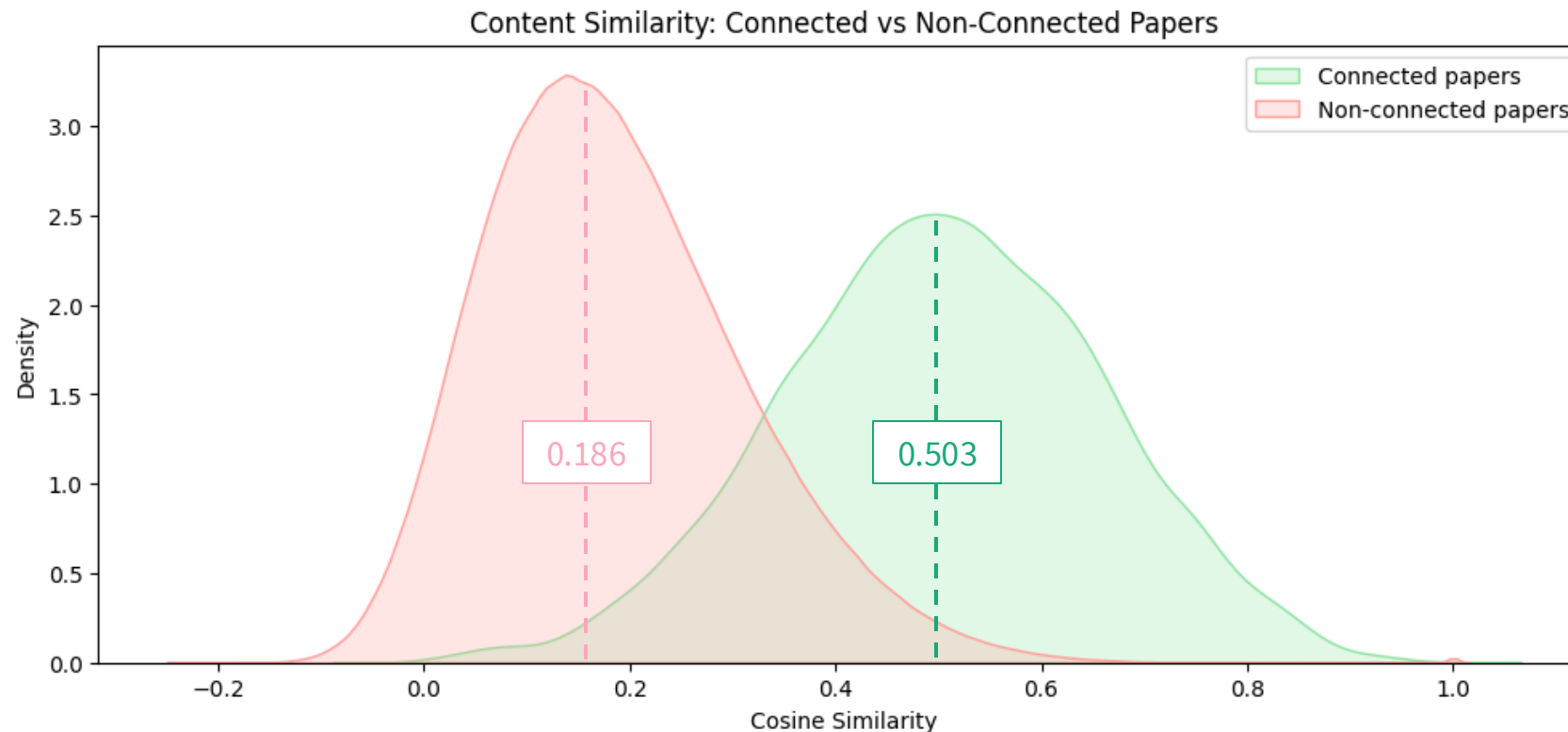
Text Embeddings

Text Embedding Quality Assessment



Text Embeddings

Graph Content Correlation Analysis



- Connected papers tend to have higher cosine similarity
- Co-cited papers may not always be similar in content (e.g., methodology, dataset etc.)

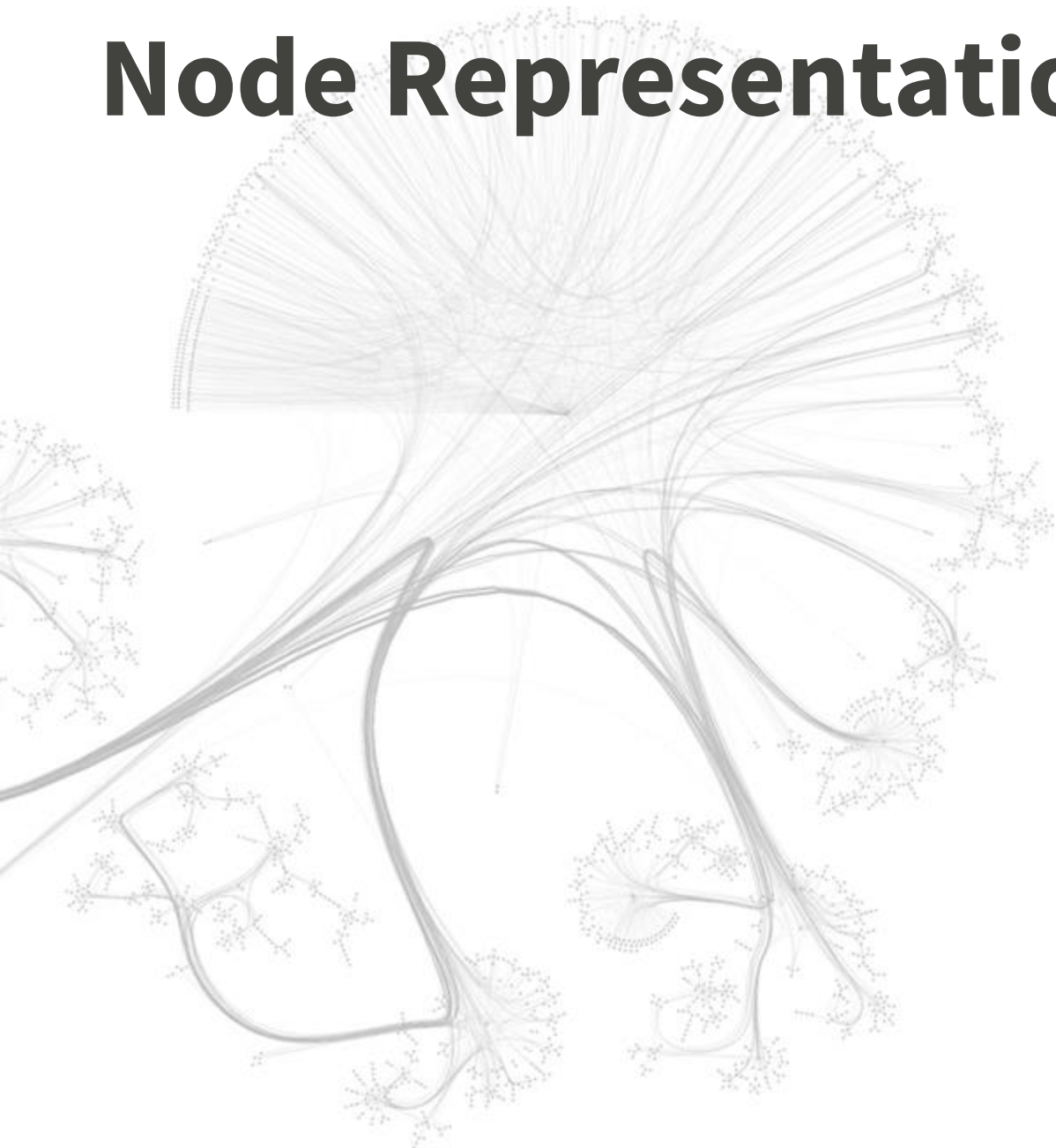
Methodology

Representation & Prediction Models

Node Representation	Link Prediction	Evaluation
Text Embeddings Neighbor Aggregation	Dot Product Scoring	Threshold setup
Text Embeddings Neighbor Aggregation + C. Measures	Logistic Regression Classifier	Evaluation Metrics
Text Embeddings + Node Embeddings	Graph Convolutional Network	Model Comparison



Node Representation



Train, Validation and Test splits



Edge Splits

Train, Validation & Test Edge Splits

	Train	Validation	Test
Split Ratio (%)	80	10	10
Positive Edges	4345	542	542
Message Passing Edges	4345	4345	4887

The three splits have the same number of nodes (**2708**)

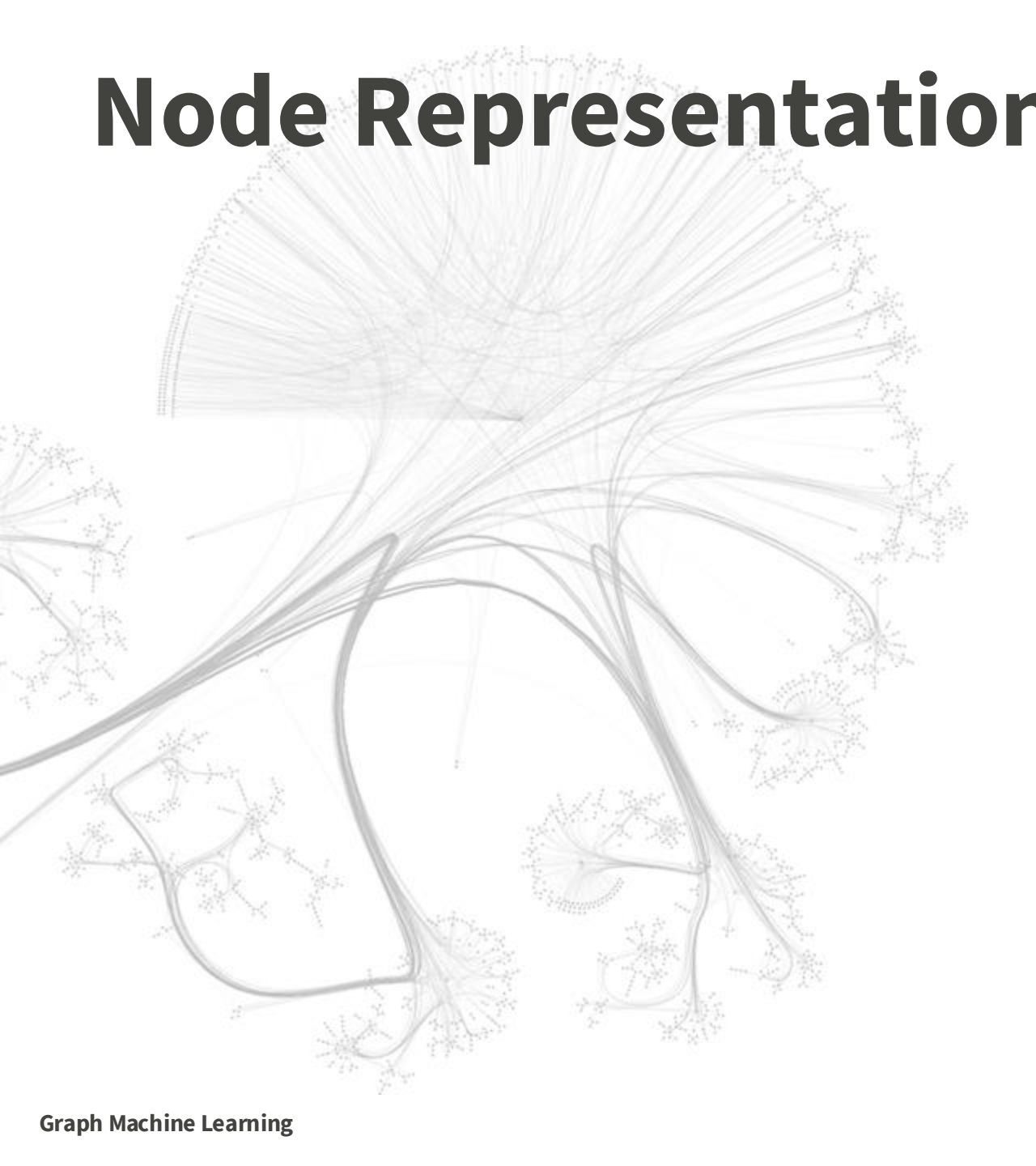
Negative splits were also generated with a **1:1** ratio

Splits generated using torch's RandomLinkSplit

*Message Passing



Node Representation



Text **Embeddings** Neighbor
Aggregation



Text Embeddings Node Aggregation

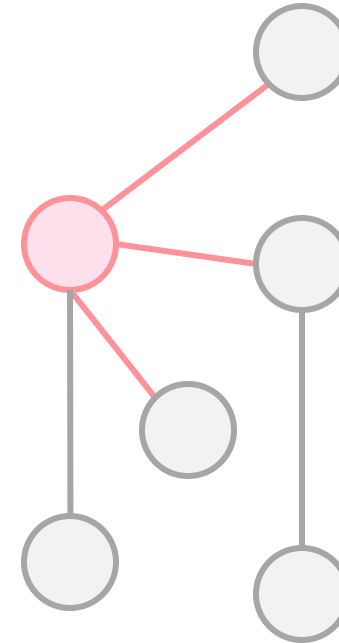
Incorporating graph structure through node aggregation

A node's representation should reflect both its own features AND the features of nodes connected to it.

How many hops? (direct and undirect neighbors).
We set $n_hops = 2$

It performs h -step random walk smoothing of node features.

Each embedding is of size 1024 (aggregation of text embeddings)



$$H = \tilde{A}^h X$$

h = number of hops

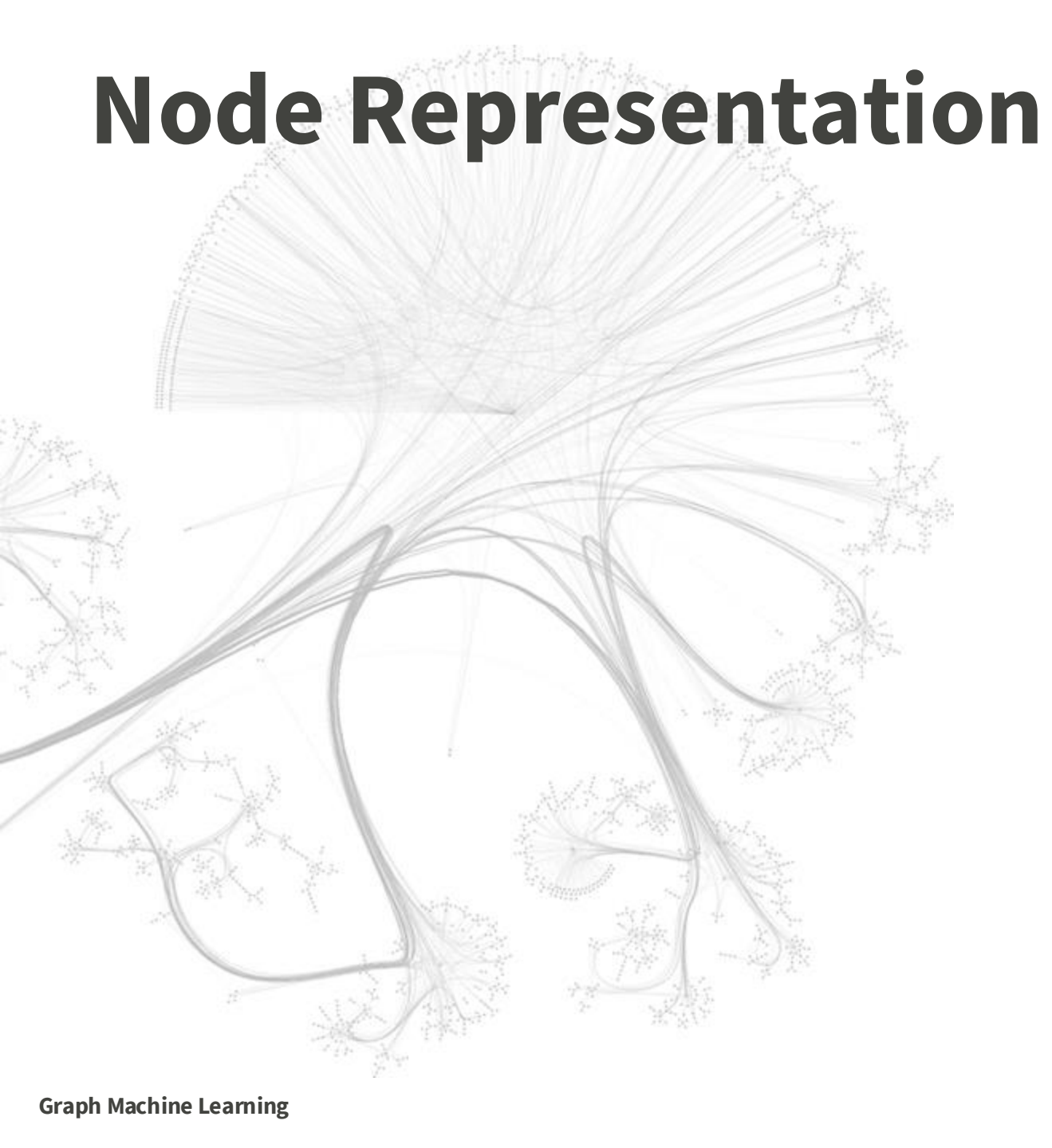
$\tilde{A}$ = Normalized adjacency matrix (w/ self loops)

X = Text embeddings

H = Node embeddings



Node Representation



Text **Embeddings** Neighbor
Aggregation
Augmented with **Centrality**
Measures



Text Embeddings Node Aggregation

Augmented with Centrality Measures

We augment the previous representation with **three** centrality measures

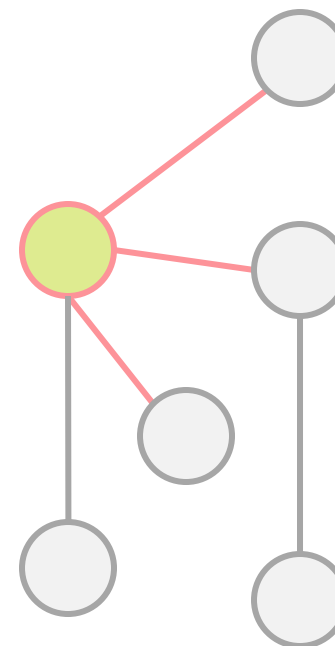
Degree: local popularity

Eigenvector: recursive influence from important neighbors

Clustering coefficient: local triangle-forming tendency

Computed on the training **message passing** graph only (to avoid leakage)

Each embedding is of size 1027 (1024 + 3 centrality measures)



$$H = \tilde{A}^h X \oplus C$$

h = number of hops

$\tilde{A}$ = Normalized adjacency matrix (w/ self loops)

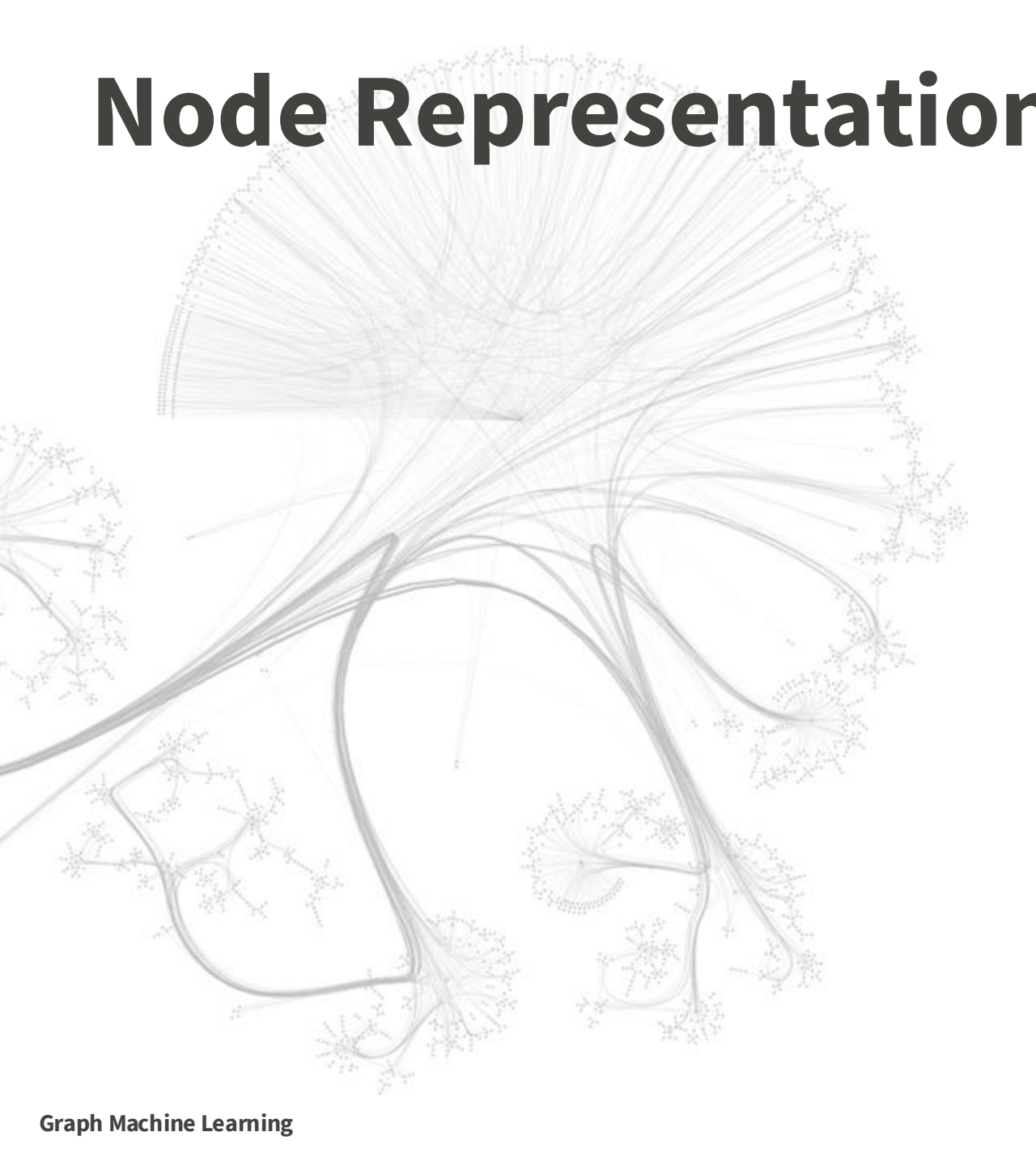
X = Text embeddings

H = Node embeddings

C = Normalized Centrality Measures



Node Representation



Text **Embeddings** +
Node2Vec **Embeddings**



Node2Vec Embeddings

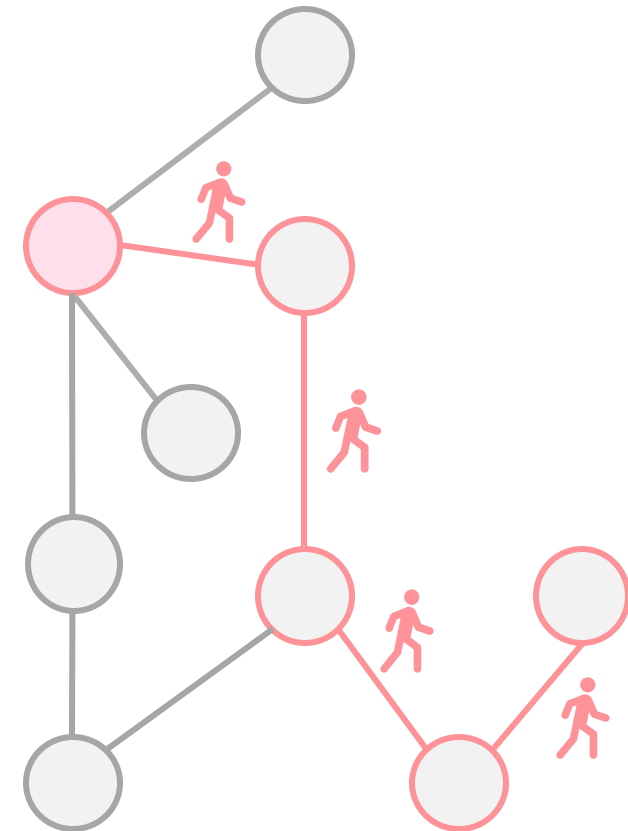
Learnable graph structure

Incorporate graph structure via Node2Vec instead of Neighbor Aggregation

We use unbiased random walk to generate 10 walks per node, 20 nodes per walk

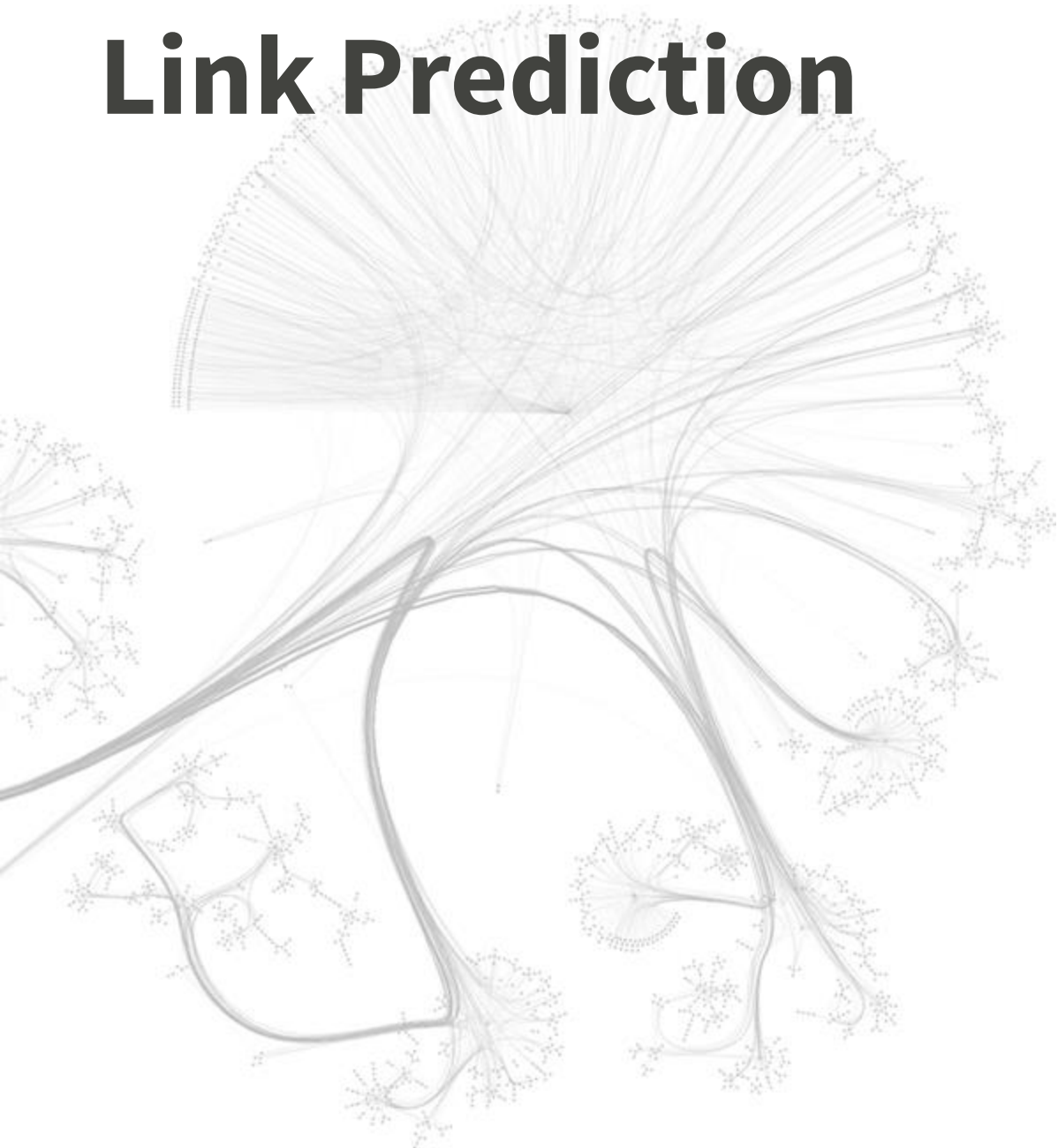
With a context size of 10 nodes and 1 negative sample per positive pair

Each embedding is of size 1152 (1024 + 128)



$$H = TE \oplus N2V$$

Link Prediction

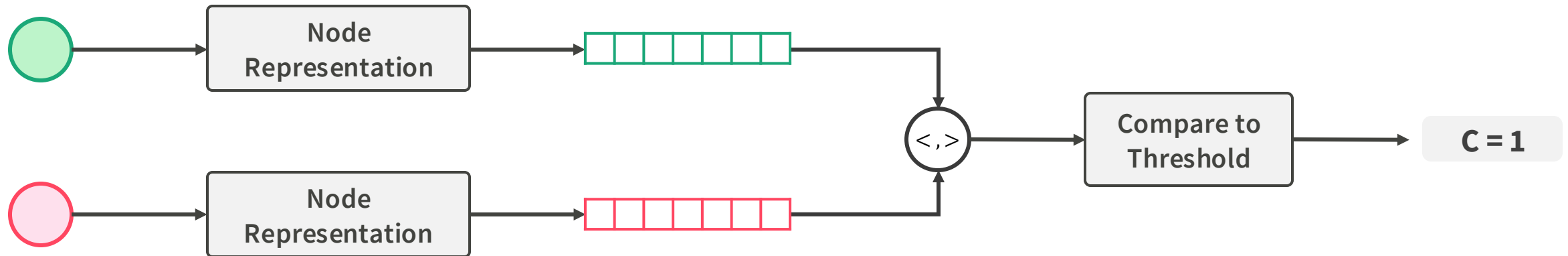


Dot Product Scoring

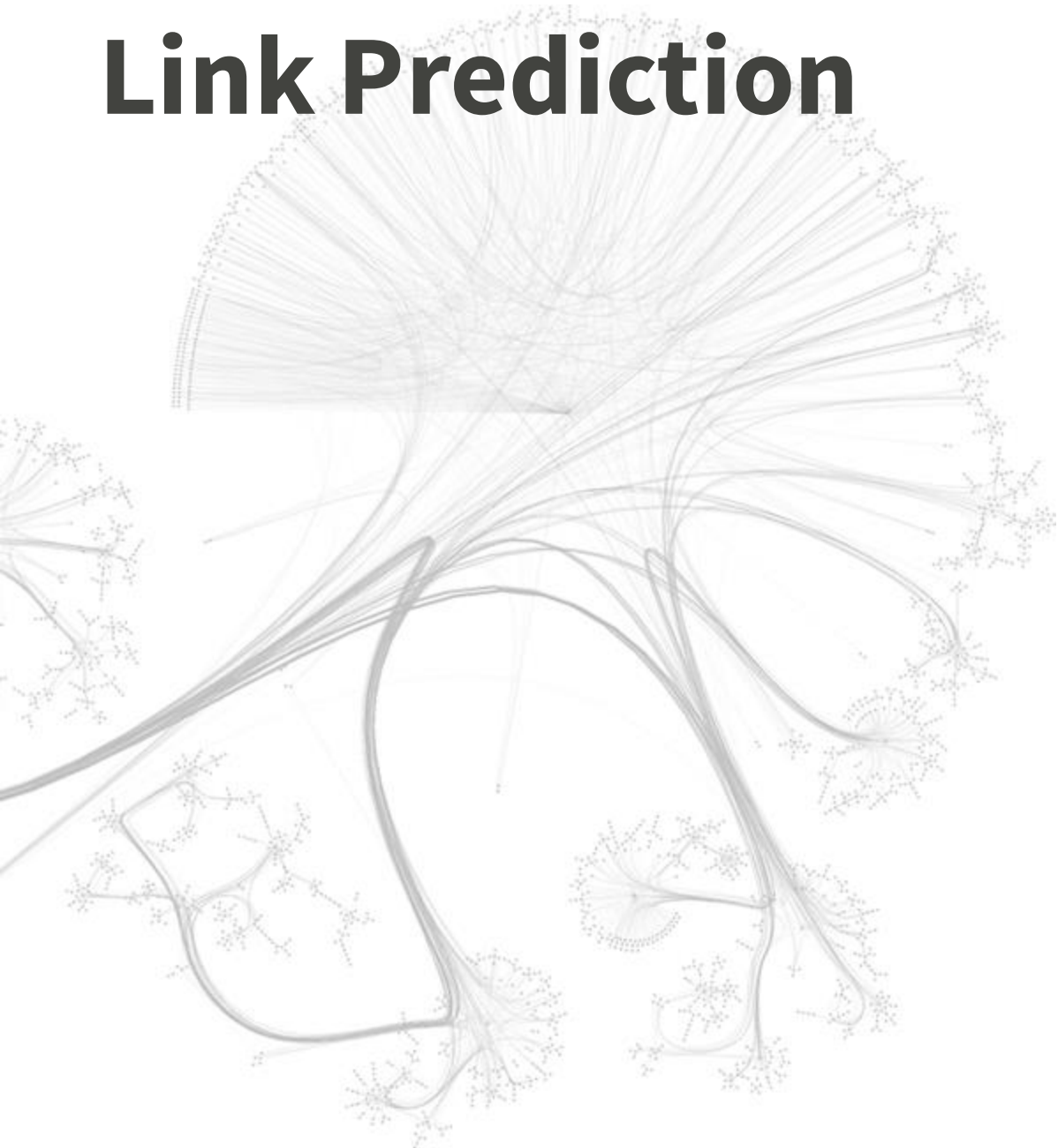


Dot Product Scoring

Measuring Similarity Scores through Dot Product Between Pair of Embeddings



Link Prediction

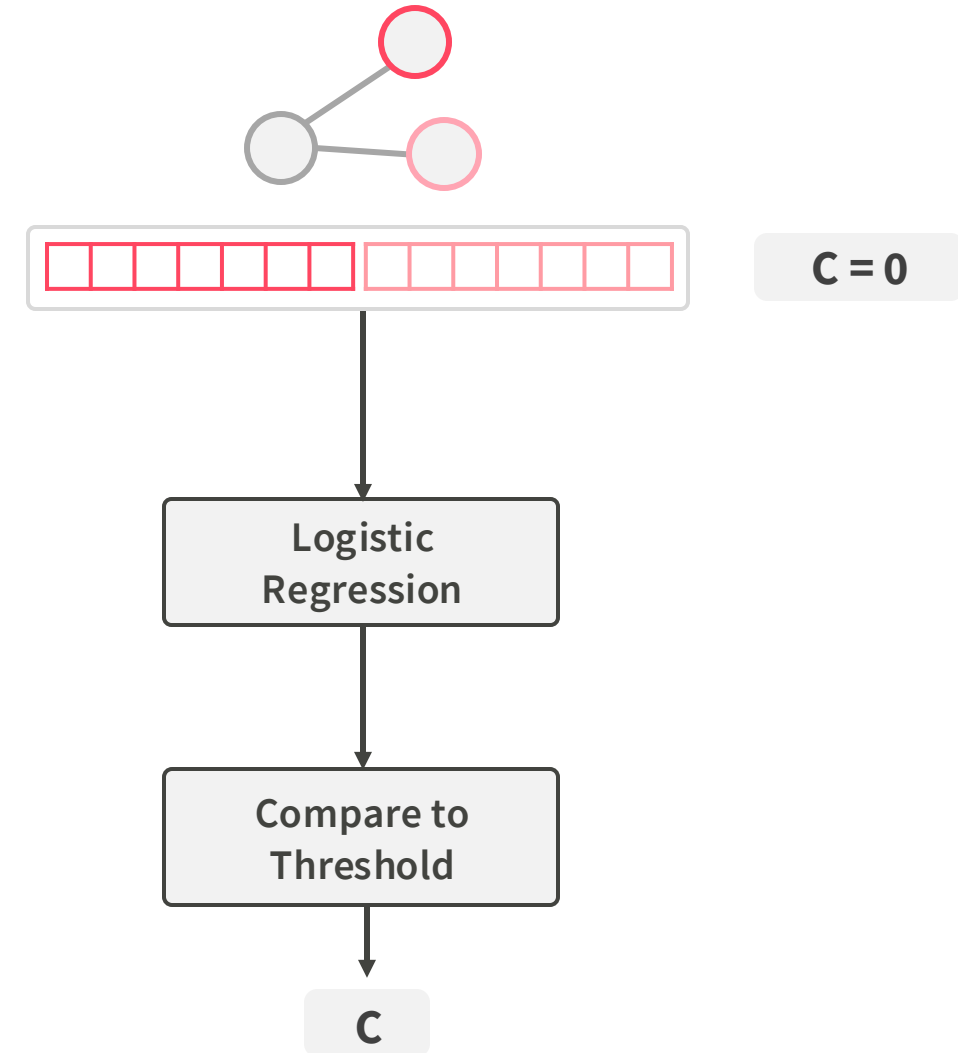
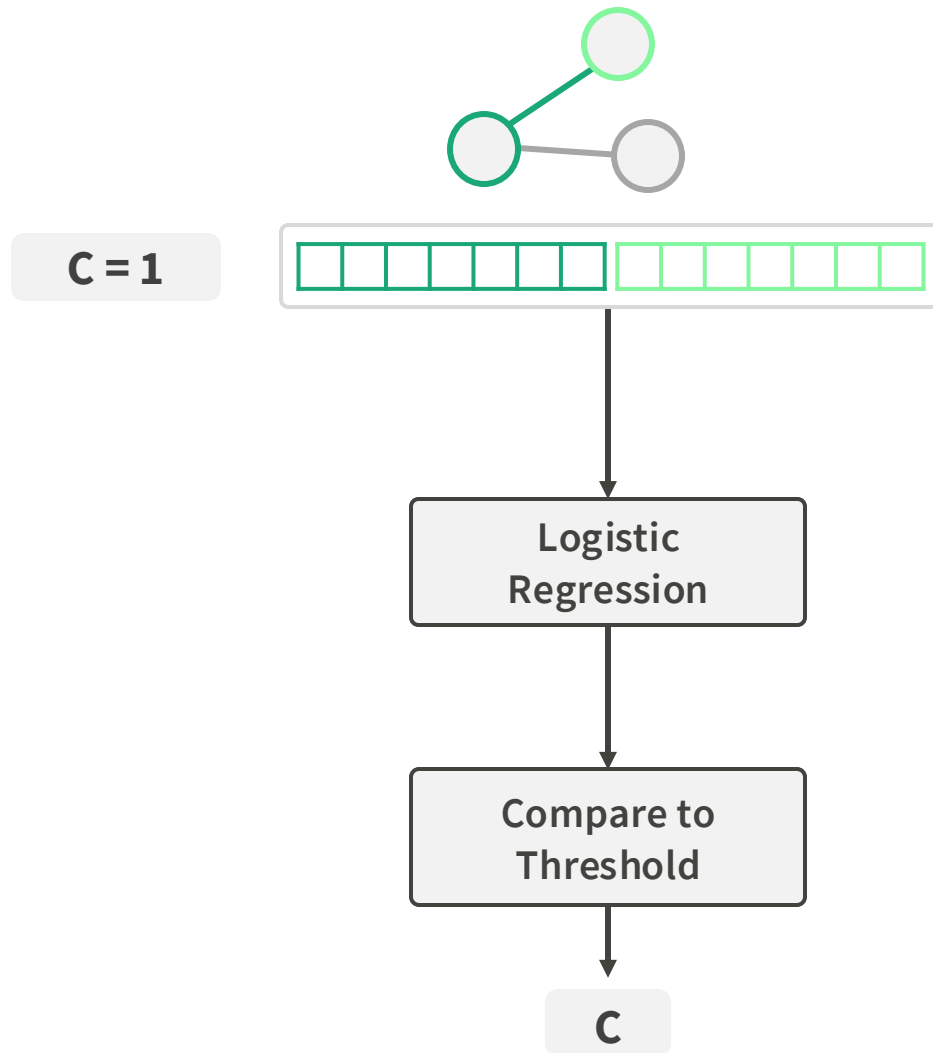


Logistic Regression **Classifier**

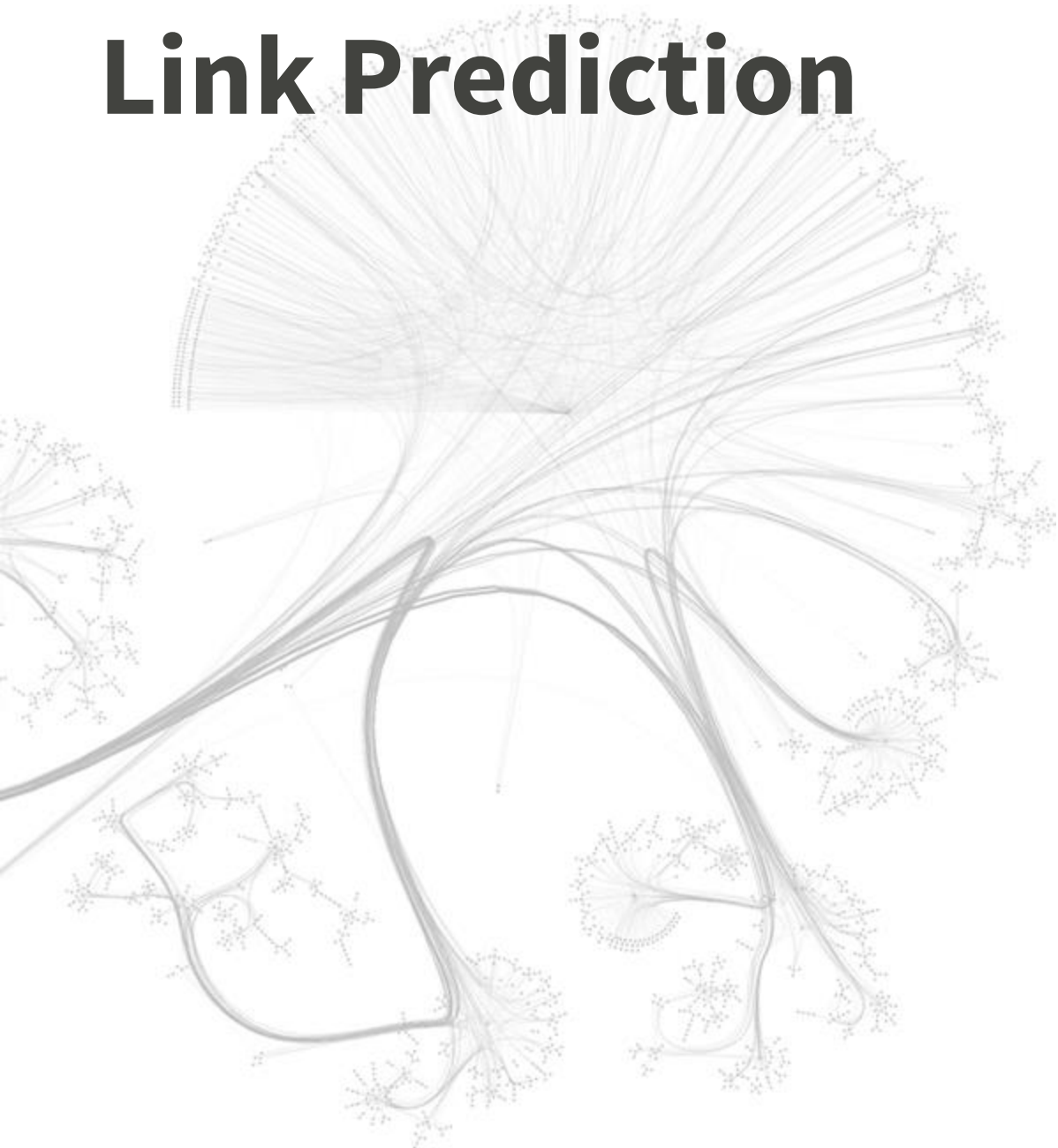


Logistic Regression

Training a Logistic Regression Classifier



Link Prediction

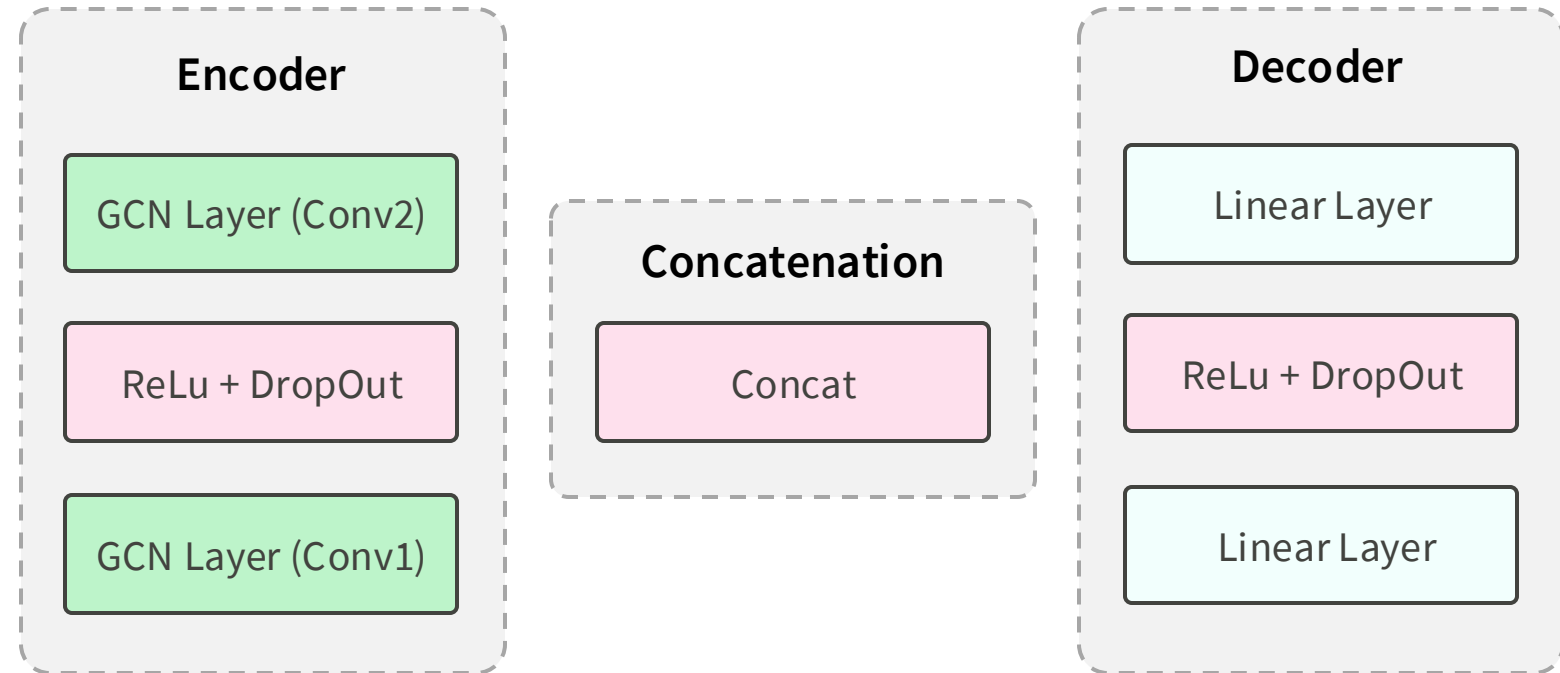
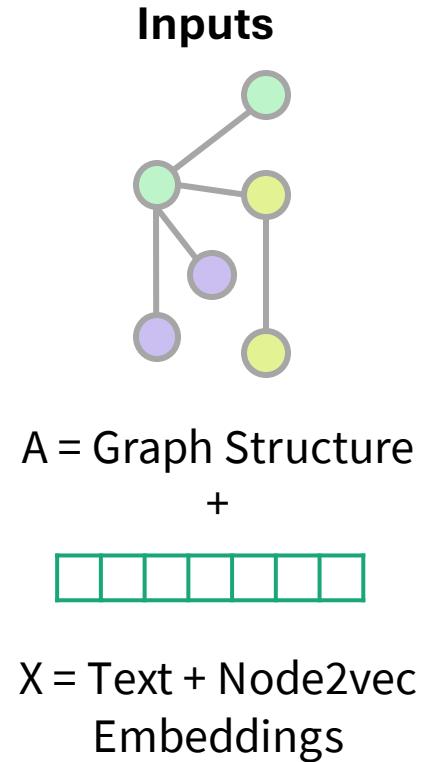


Graph Convolutional Network
(GCN)

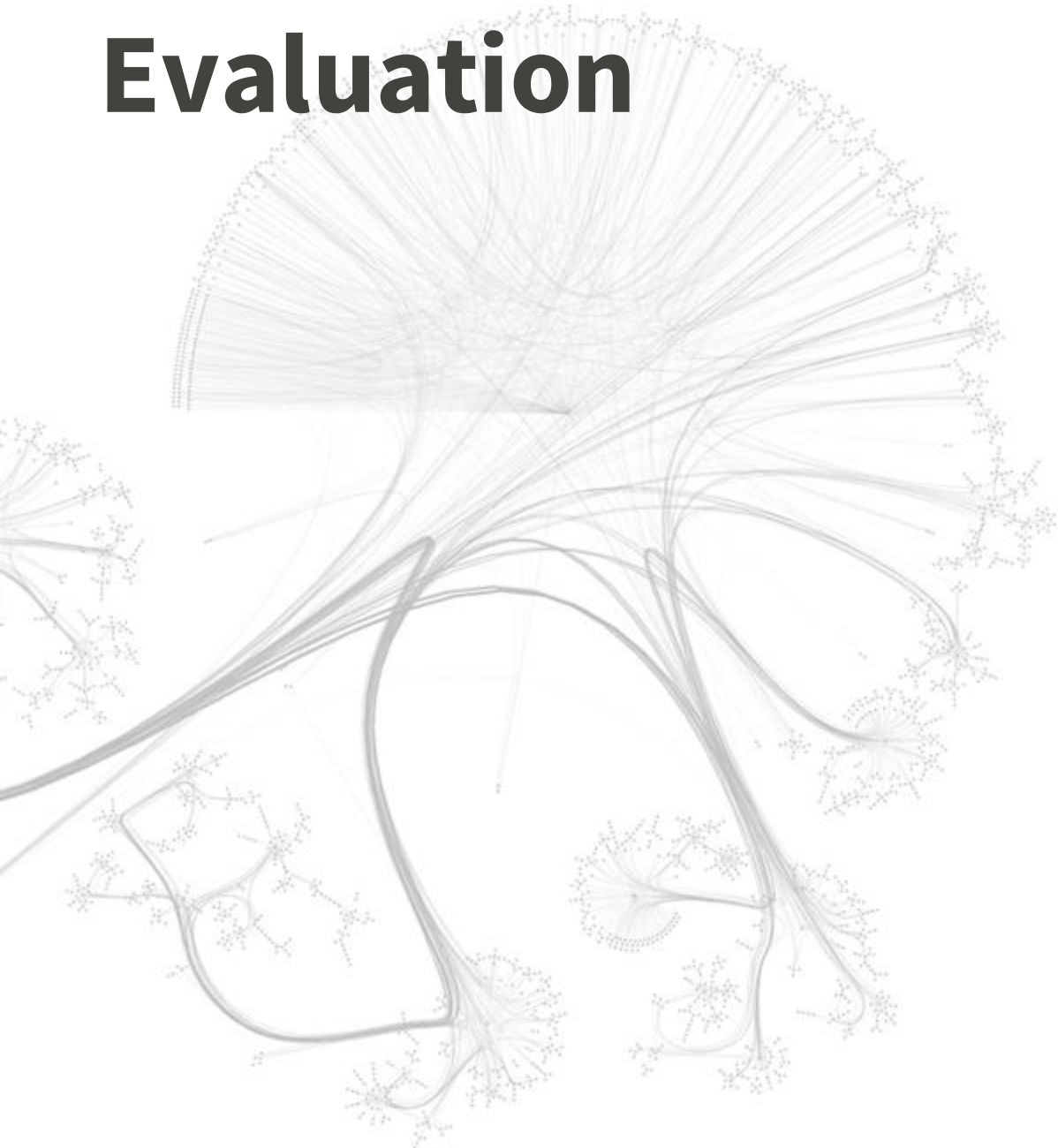


GCN

Using a Graph Convolutional Network



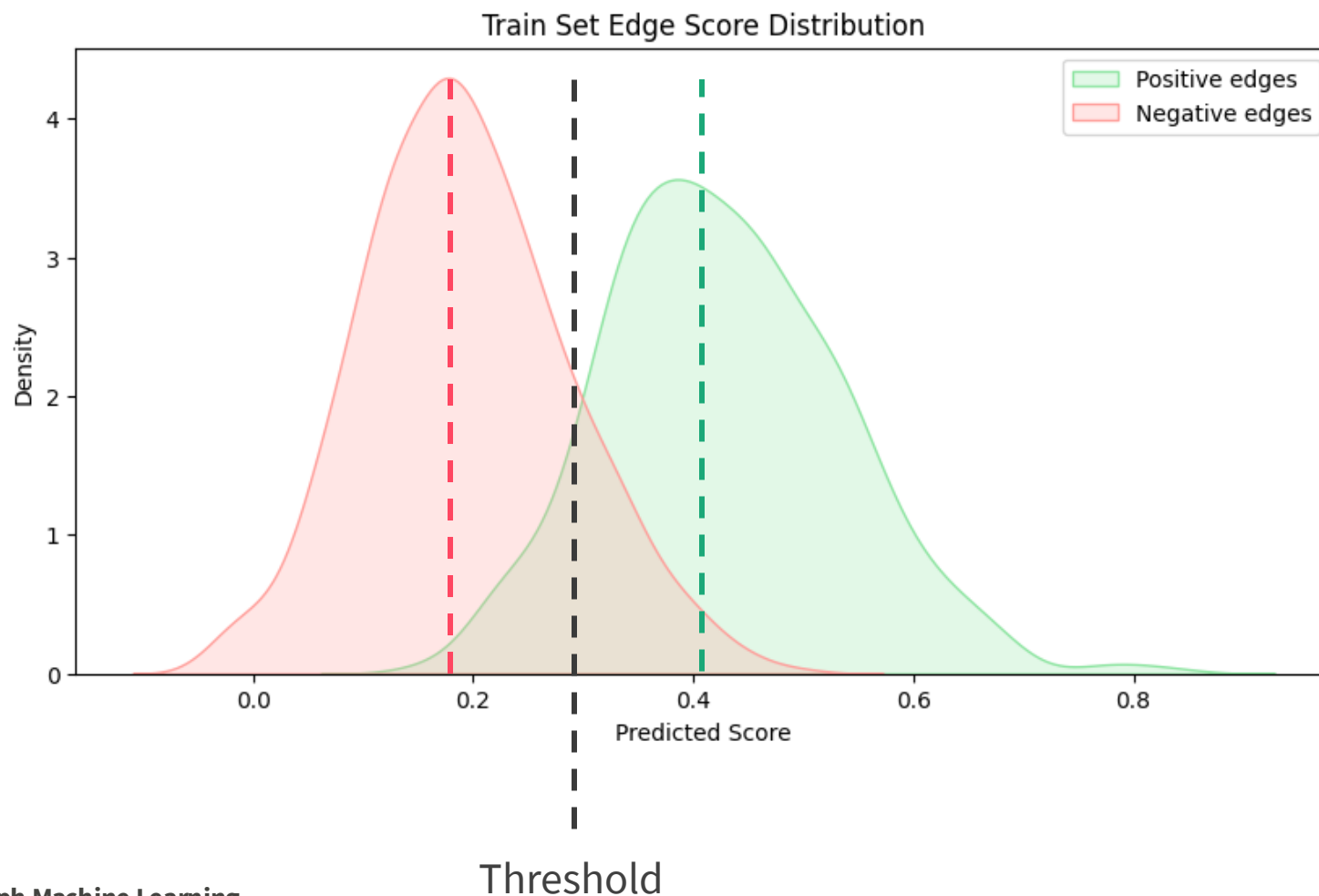
Evaluation



Fixing a **Classification** threshold

Classification Threshold

Dynamic definition of classification thresholds



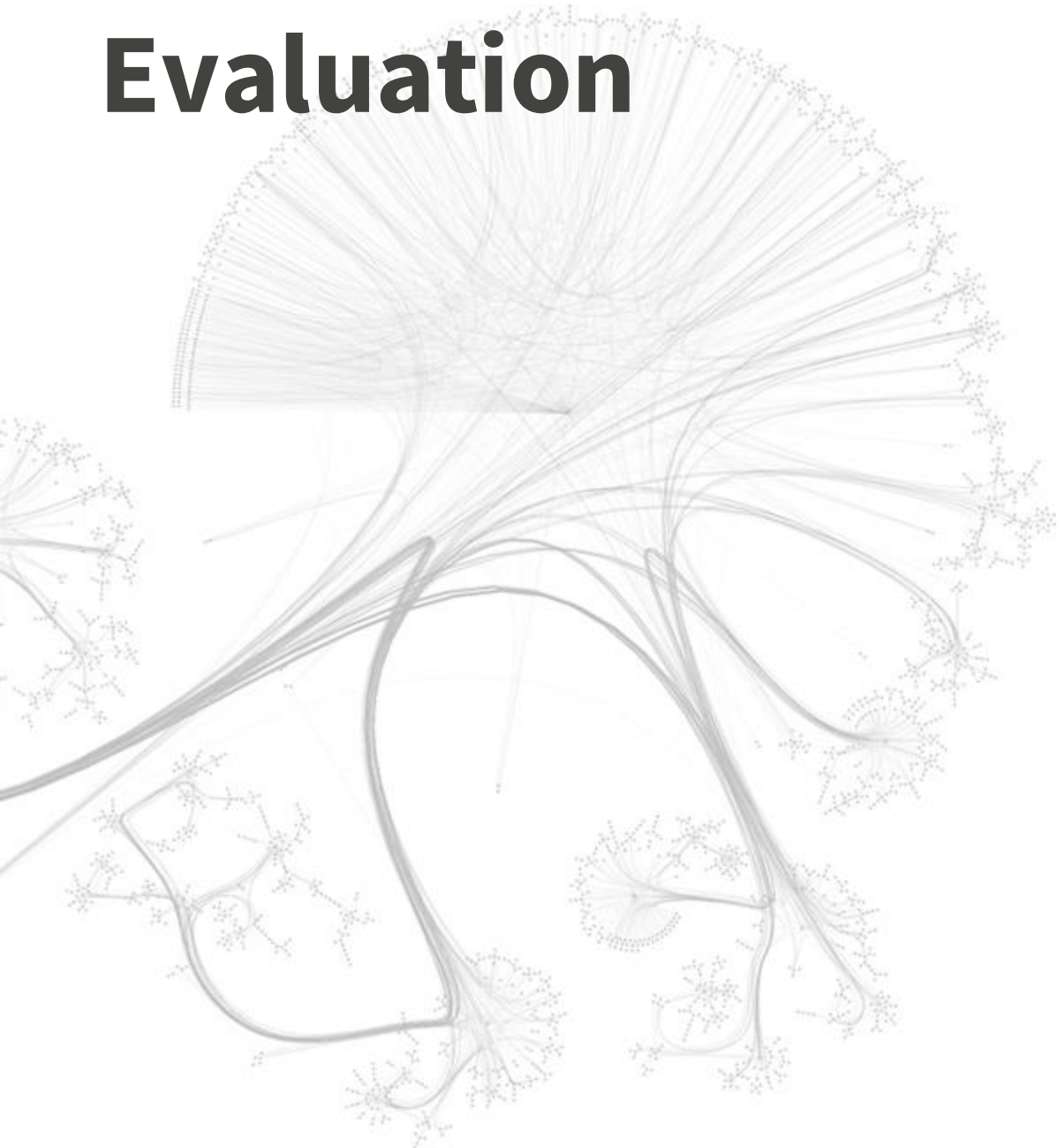
How to choose the best similarity threshold?

Score distributions separation on train set

Threshold = Means difference



Evaluation



Choosing the right classification
metrics



Classification Metrics

Dynamic definition of classification thresholds



Separability Power

ROC - AUC



Prediction Quality

Average Precision AP

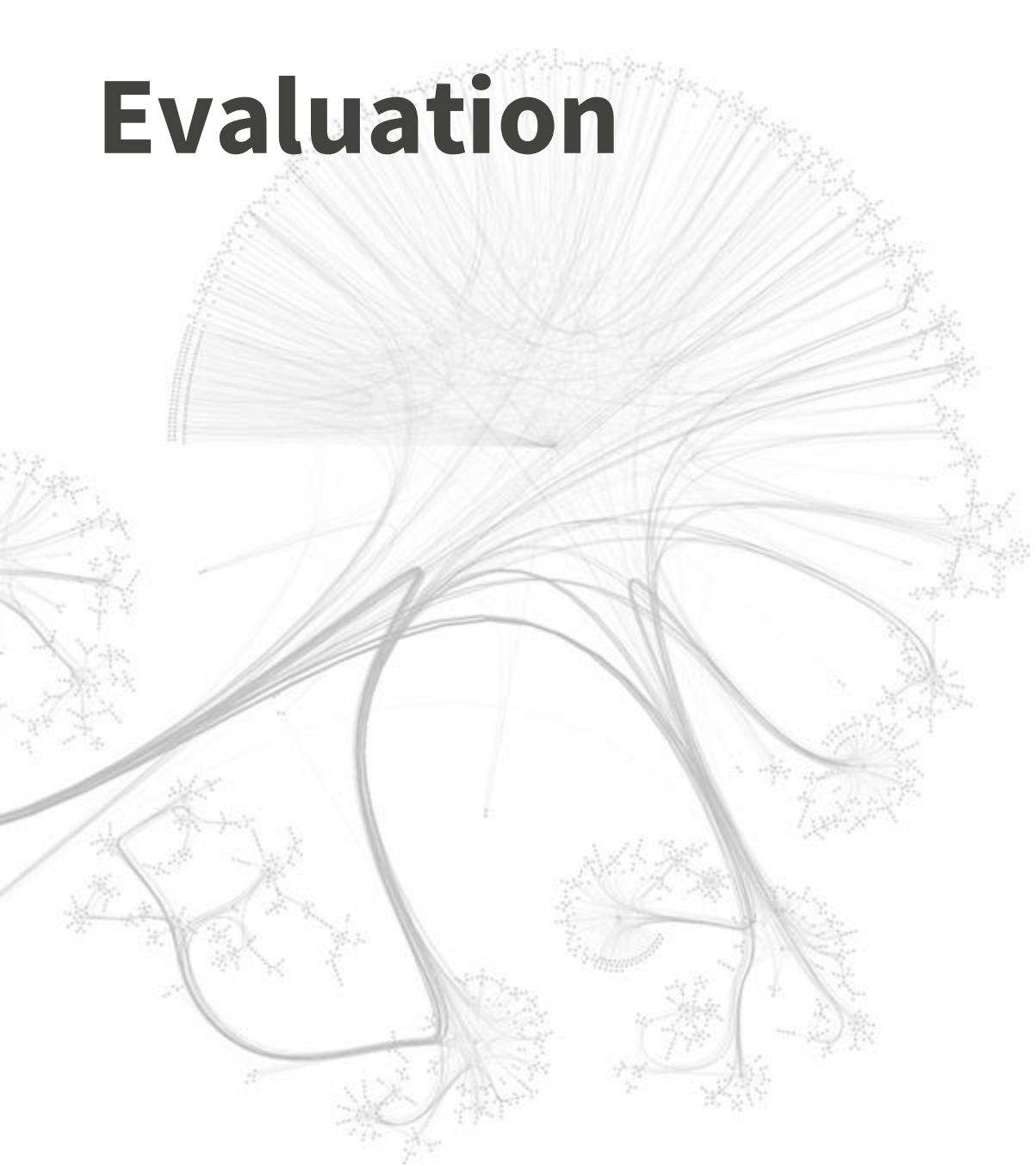


Decision Reliability

Accuracy



Evaluation



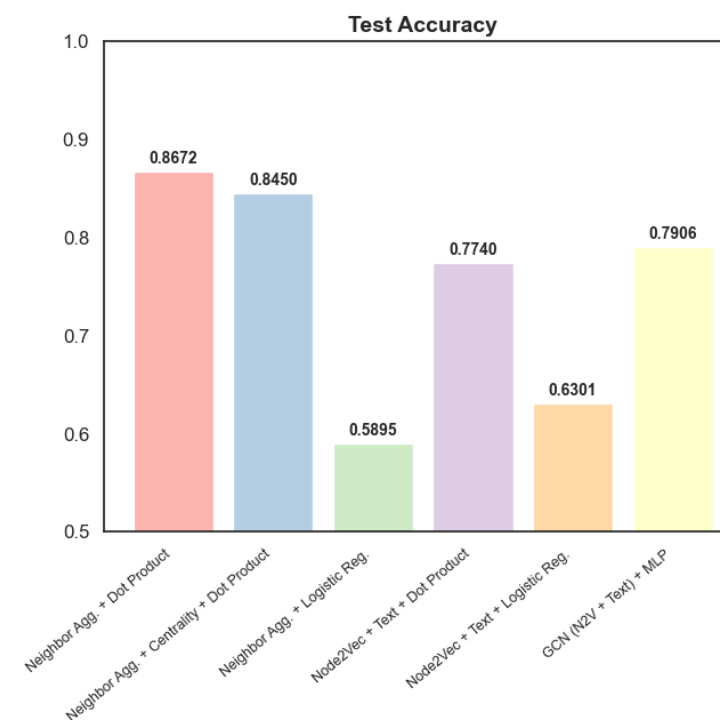
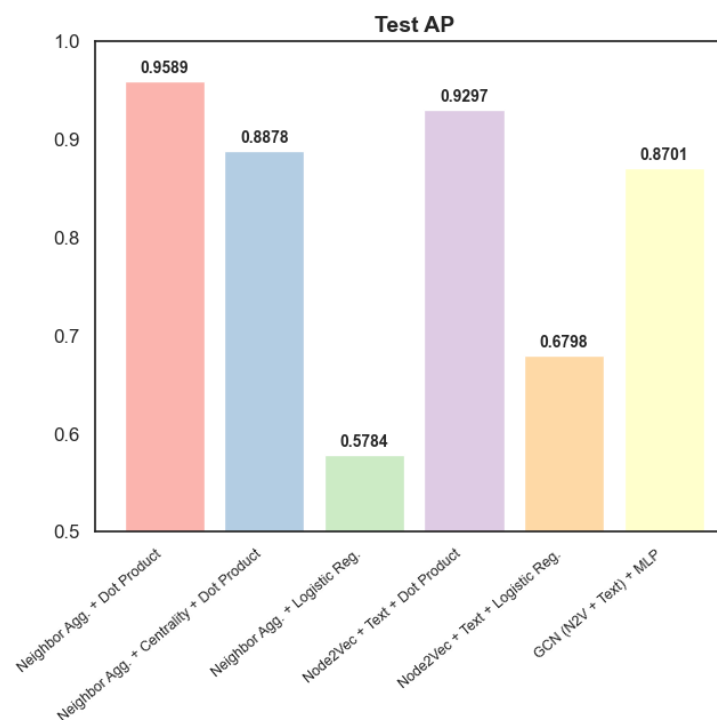
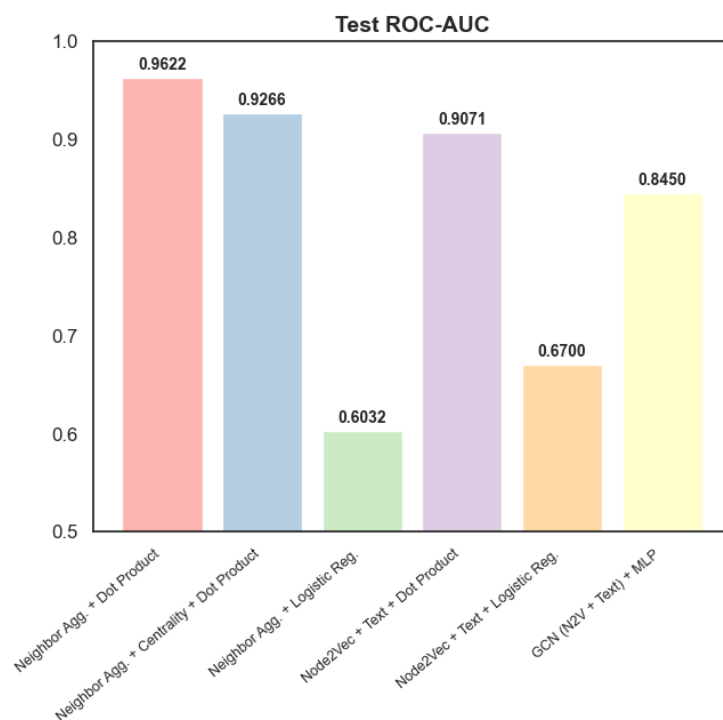
Results & Model Comparison



Test Set Results

Model Comparison

Link Prediction: Final Performance Comparison



Interpretation

& Final Decision

Simplicity wins on this task

GCN might be an overkill for this task (with this quality of attributes)

Node2Vec incorporate learnable graph structure but doesn't outperform neighbor aggregation (hyperparameter tuning?)

Logistic Regression fails with all types of embeddings



Text Embeddings Neighbor Aggregation with Dot Product Scoring



Thank **you** for
your **attention**

